Strengthening the Rule of Law through Community Policing: Evidence from Liberia[*]

Benjamin S. Morse[†]

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[†]Technical Director, Social Impact bensmorse@alum.mit.edu

Abstract

How to improve security and strengthen the rule of law in fragile states? Community policing programs have long been at the forefront of policymakers' efforts to address this challenge. These programs tend to be more expansive than those found in developed countries, focusing not only on building trust through meetings and foot patrols, but also on eliciting 'co-production' from communities to supplement scarce police capacity and provide alternatives to vigilantism. I partnered with the Liberian National Police (LNP) to experimentally evaluate the effectiveness of this approach in Monrovia, Liberia, one of sub-Saharan Africa's most crime-ridden cities. Drawing on a large-scale resident survey and administrative crime data, I find that the program improved relations between police and citizens, strengthened social norms against vigilantism, and mobilized communities to participate in the police's "Watch Forum" initiative by forming and sustaining local security groups designed to facilitate cooperation with police. These changes were accompanied by a roughly 40 percent reduction in the incidence of mob violence. Despite these improvements, the program did not reduce the overall incident of crime, improve perceptions of security, or increase crime reporting.

1 INTRODUCTION

Insecurity due to crime and violence is pervasive in many of the world's weakest states (Mack et al., 2013). To address this problem, donors and aid organizations invest millions of dollars annually in assistance programs designed to strengthen the capacity of state security forces and legitimate their authority in the eyes of citizens. This two-pronged approach has long dominated the practice of security sector reform, and is motivated by the idea that institutional capacity and citizen cooperation are "close complements" in the production of security and therefore must be built in tandem if countries are to escape fragility and violence (Baranyi, Beaudet and Locher, 2011).

Yet in recent years a growing number of scholars have begun to question whether this narrow, state-centric approach to security sector reform is optimal, and in particular, whether it adequately addresses short term security needs (Baker, 2009; Chirayath, Sage and Woolcock, 2005; Wisler and Onwudiwe, 2008). At the core of these concerns is a fundamental problem of sequencing: building strong, effective security forces takes decades; in the interim, encouraging citizens to rely on these institutions even as they remain fundamentally untrustworthy and unreliable is unlikely to lead to meaningful improvements in security, and could potentially backfire if engaging with corrupt and incompetent security forces serves to harden citizens' negative perceptions.

Traditional, state-centric models of security reform also struggle to provide a viable alternative to extrajudicial practices such as mob violence, lynching, and vigilantism. Instead, they address these problems with outreach and awareness programs designed to foster norms that support the rule of law and reject extrajudicial practices (Carothers, 2006). Yet because these interventions do little to address the deficiencies of state security forces and little to reduce citizens' security concerns, reliance on extrajudicial mechanisms tends to persist.

If building effective, legitimate security institutions takes decades, how can policymakers improve security and reduce reliance on extrajudicial practices in the short term? In this paper, I partner with the Liberian National Police (LNP) to test an approach rooted in "multidimensional" models of security reform, which call for policymakers to recognize that multiple non-state actors often play a role in security provision in countries with severe and deeply-rooted capacity deficits (OECD, 2008; Baker and Scheye, 2007). In these settings, "first-best" solutions founded on the Weberian ideal of the state as the sole legitimate provider of security are not always possible; instead, the best way to "respond to the short-term needs of enhanced security and justice service delivery, while also building the medium-term needs of state capacity" may be for security forces to develop "productive partnerships" with local communities and non-state security actors (OECD, 2008, 7).

The challenge, of course, is to do so in a way "that builds on local networks and institutions *without* encouraging vigilantism" (Rose-Ackerman, 2004, 187, emphasis mine). Liberia's model of community policing responds to this challenge by expanding the scope of traditional, 'Weberian-style' community policing programs, to focus not only on encouraging reliance on the police, but also on providing communities with a lawful, rights-respecting avenue through which they may 'coproduce' certain security functions in coordination with the police.

Coproductive models of community policing are increasingly common in fragile states, and often receive considerable support from aid agencies and international donors (Wisler and Onwudiwe, 2008; Dinnen and Peake, 2013; Kagoro, 2019). But there is little rigorous experimental evidence to inform whether these programs are effective. To address this gap, I partnered with the Liberian National Police to experimentally evaluate the effectiveness of their community policing program in Monrovia, one of sub-Saharan Africa's most crime-ridden cities and a place where authorities have long struggled to combat vigilantism and mob violence (Zanker, 2017; Downie, 2013). Working with officers across nine of Monrovia's ten police zones, I identified a sample of 93 communities that were eligible for the program, and randomly selected 45 of these communities to receive the program for a period of 10 months, from February to November 2018.

Using survey data, official crime reports, and qualitative field reports from research assistants assigned to shadow the police for the duration of the study, I provide four sets of results. First, I find that the program improved relations between police and citizens, making residents of treatment communities more likely to view the police was capable and well-intentioned, and more likely to personally know individual officers. Second, and more significantly, the program mobilized communities to more actively organize, support, and sustain local security groups. While such groups were common in control communities as well, groups in treatment communities were more likely to be registered with the police's Watch Forum initiative and were sustained by greater contributions of time and effort from residents.

A central concern for policymakers is whether coproduction models of community policing can re-orient communities' efforts to self-provide security towards more rights-respecting practices. My third set of findings strongly suggests this was the case. In addition to being more trusting of police, residents of treatment communities were more knowledgeable about the rules and guidelines governing local security groups (especially with regards to suspects' rights) and less willing to condone mob violence. Finally, and perhaps most significantly, residents of treatment communities reported 40% fewer instances of mob violence in the year preceding the survey, as compared to residents of control communities. This result is replicated using mob violence event data from an independent survey of police officers not directly involved in the study.

Although the program made communities more trusting, better organized, and less supportive of mob violence, these improvements did not lead to reductions in the overall incidence of crime or the incidence of any particular category of crime, as measured by both the survey and crime reports from the police. The program also did not increase crime reporting or individual forms of cooperation with police, such as the provision of crime tips or assistance with police investigations. And it did not improve residents' perceptions of security or their overall satisfaction with police performance.

These null effects suggest that policymakers should remain clear-eyed about what community policing programs can achieve in countries like Liberia. They are not a solution to broader problems of crime, violence and police (in)capacity in fragile states, and they cannot "short-circuit" the long process of security sector reform by mobilizing coproduction from citizens. So long as the police remain incapable of responding effectively to crimes or thoroughly investigating cases, residents are likely to remain insecure, reluctant to report crimes, and unsatisfied with police performance.

This study makes two main contributions to the academic literature. First, this study contributes to the state-building literature on governance in "hybrid political orders" by showing that delegation to local communities need not undermine state authority, and may in some circumstances strengthen it by diverting communities away from extralegal practices (Boege, Brown and Clements, 2009).[1] This conclusion runs counter to traditional perspectives on state-building, which have long viewed hybrid spaces as a realm of conflict in which state and non-state actors confront each other in a zero-sum competition for social control (Migdal, Kohli and Shue, 1994, 66). The findings reported here, by contrast, draw attention to the many forms of collaboration and accommodation that occur along the path to state centralization, and the role that these arrangements can play in strengthening the state and its rule of law in the short term.

This study also contributes to the literature on the causes of vigilantism and mob violence. Whereas scholars have historically viewed these practices as the "best response" of communities plagued by high levels of crime and weak or absent security institutions (Abrahams, 1998; Jung and Cohen, 2020; Rotberg, 2004), more recent research contests this view, arguing that violent, extralegal practices in fact do little to improve security, and are instead driven by emotive reactions to crime, such as anger, moral outrage, and desire for retribution (García-Ponce, Young and Zeitzoff, 2018; Darley, Carlsmith and Robinson, 2000; Johnson, 2009). Mob violence is irrational, according to this perspective, and there is little policymakers can do to combat it. My findings are inconsistent with this view, and instead suggest that coproductive models of community policing can redefine what communities view as their "best response" to crime in settings plagued by weak policing.

[1] Following Boege, Brown and Clements (2009) and much of the literature on multi-layered models of security reform (Dinnen and Peake, 2013), I use the term *hybrid* to refer to security systems in which both state and non-state actors contribute to the production of security.

2 Community Policing in Fragile States

Community policing programs are premised on the idea that citizen cooperation is essential for effective policing (Moore, 1992; Tyler and Fagan, 2008). Because police cannot be everywhere all of the time, they must rely on citizens to learn about crimes that have occurred, identify potential suspects, and gather evidence to bring wrongdoers to justice. Recognizing the central role that citizens play in crime prevention, proponents of community policing have called for a shift in how police work is done, from strategies oriented around the exercise of coercive law enforcement power, to strategies focused on building and sustaining relationships with citizens, community leaders, and civil society organizations (Gill et al., 2014). The hope is that as residents' confidence and familiarity with the police improves, they will become more cooperative and more willing to provide information to the police. And as the police become more familiar with the communities they serve, they will tailor their strategies to more effectively address the drivers of crime. Together, greater cooperation from citizens and greater local knowledge among rank and file officers will improve the police's overall ability to prevent, detect, and solve crimes (Ferreira, 1996).

Although community policing programs originally came to prominence in the U.S. and UK during the 1970s and 1980s, in recent years they have become popular in fragile and conflict-affected states as policymakers search for cost-effective strategies to improve police legitimacy and strengthen the rule of law. Major donors such as the World Bank, DFID, USAID and others now spend millions of dollars annually to promote community-oriented policing programs (Denney and Jenkins, 2013). The UN, for its part, has made community policing a mainstay of its peacekeeping strategy, calling for "community-oriented policing and intelligence-led policing to guide all operational activities of the United Nations police in their support to host state police" (United Nations, 2016, 7).

To a significant degree, support for community policing in fragile states is based on the same premise that undergirds support in developed countries — namely, that greater cooperation from citizens will lead to greater effectiveness on behalf of the police. The UN's manual on community policing, for example, describes the logic behind community policing as follows: "greater public trust and confidence in the police leads to an enhanced flow of quality information from the public, which in turn fosters increased police effectiveness" (United Nations, 2018, 10). And greater police effectiveness, in turn, creates a more positive public perception of the police, precipitating a "virtuous cycle" of trust and police effectiveness (United Nations, 2018, 11).

Yet scholars have increasingly begun to question whether this optimistic actually logic holds in fragile, low-capacity states, where security forces often face shortages of personnel, vehicles, and even basic supplies such as batons, radios and stationary (Downie, 2013; Baker, 2009). From a theoretical perspective, the same premise that motivates community policing — that citizen cooperation and police capacity are 'close complements' in the production of security — also implies that greater cooperation will be of little benefit when the police cannot reliably respond to incidents, investigate reported crimes, or follow-up on crime tips. Absent broader, more comprehensive reforms addressing police capacity constraints, community policing programs may be bound to fail.

There are other risks to community policing in these settings as well, beyond the risk of failure. The irony of promoting reliance on police who remain fundamentally unreliable is not likely to be lost on ordinary citizens, whose everyday interactions with the police often involve acts of incompetence or corruption — experiences which research suggests tend to have an outsize influence on citizens' perceptions of the police (Sahin et al., 2017; Brunson, 2007). Positive interactions during town hall meetings and foot patrols are unlikely to outweigh these experiences, and may be ignored altogether if citizens view these overtures as "cheap talk", with potentially lasting damage to police credibility (Steinberg, 2008, 36). Although research on the short-term impacts of police-community engagement in areas of limited statehood has found positive impacts on perceptions (Karim, 2020), research on the medium-term impacts finds no effect, consistent with the "cheap talk" hypothesis (Blair, Karim and Morse, 2019).

In addition to concerns about the ability of the police to follow-through on their promises, observers have expressed concerns about the fact that traditional models of community policing tend to view security in the Weberian sense, as “a non-negotiable state monopoly” (Wisler and Onwudiwe, 2008, 435), when in reality communities pursue security through a variety of avenues, ranging from private security firms, to neighborhood watch forums, to vigilante groups, to mob violence (Tapscott, 2021; Rose-Ackerman, 2004). Though these actors operate outside the law, they are often supported by ordinary citizens, and they often serve as important providers of security, deterring crimes when the police are too weak or ineffective to do so themselves (Baker, 2009). In such settings, encouraging reliance on the police at the expense of local sources of security could exacerbate problems of crime and violence if the police prove to be ‘imperfect substitutes’ for non-state security providers, with potentially negative consequences for long-term efforts to promote reliance on formal-sector justice and security institutions.

2.1 Community policing and ‘multi-layered’ models of security reform

Concerns about the viability of community policing programs in fragile states are emblematic of what Andrews, Woolcock and Pritchett (2017) refer to as the problem of “premature load bearing” in the field of development, which occurs when “institutions and organizations are required to perform tasks before they are actually capable of doing them” (Andrews, Woolcock and Pritchett, 2017, 54). The result, they argue, is akin to “putting too much weight on a structure before it is able to support it ... not only does this not accomplish the task at hand, it also sets progress back” (54).

Premature load bearing is common in many sectors of states undergoing reform, but it may be especially common in the justice and security sectors because of the premium placed on the state as the *sole* legitimate provider of these services. In line with Weberian notions of state authority, efforts to improve security and strengthen the rule of law typically focus exclusively on *state* security institutions, and very much depend on citizen cooperation and reliance for success. Yet because these reforms take decades to materialize, policymakers often find themselves in the uncomfortable position of promoting reliance on security institutions which are still very much in the process of reform.

This contradiction is not lost on policymakers, however, and in recent years many have begun to advocate for a more "multi-layered" approach to security sector reform that builds on local networks and institutions in order to supplement police capacity and ease the demands placed on under-resourced police (Baker and Scheye, 2007; Chirayath, Sage and Woolcock, 2005; Denney, 2014). This approach calls for policymakers to recognize that "first-best" solutions involving state delivery of security and justice services may be unrealistic when the state faces significant capacity deficits and lacks legitimacy in the eyes of large segments of its population (OECD, 2008, 7). Rather than focus exclusively on state institutions, reforms should focus on developing partnerships between police and local leaders and communities, and empowering these actors to contribute to the provision of their own security within the confines of established rules and guidelines. In this way, the multi-layered approach to security reform aims to direct the collective action potential of communities towards activities which "complement rather than undermine the state's ability to provide security," and thereby help "respond to the short-term needs of enhanced security and justice service delivery, while also building the medium-term needs of state capacity" (OECD, 2008, 11).

In practice, multi-layered security reforms require a degree of the delegation on behalf of the police, as well as a corresponding amount of 'coproduction' by communities (Wisler and Onwudiwe, 2008; Ostrom, 1996). For example, police may delegate authority over misdemeanor crimes to local leaders or chiefs, encouraging them to adjudicate these cases while referring more serious ones to the police solving crimes. Similarly, police may delegate responsibility for night-time vigilance patrols to neighborhood watch groups, freeing officers to focus on responding to crimes and incidents. In both of these arrangements, inputs from government are combined with complementary inputs from "citizen producers," resulting in an improvement over government production alone (Ostrom, 1996, 1082).

For those involved in tailoring traditional models of community policing to meet the needs of fragile states, the concept of coproduced policing has been influential (Baker, 2009; United

Nations, 2018). Increasingly, these programs focus not only on building trust and compliance with the police, but also on encouraging communities to form coproductive partnerships with police. In urban settings, one of the principal ways that this approach has been put into practice is through community watch programs, wherein communities elect volunteers who, after being trained and vetted by the police, are tasked with keeping watch over their communities at night, assisting with police investigations, and providing information about potential criminal activity. More generally, members of these groups may serve as advocates for the police in their communities, encouraging their fellow citizens to remain vigilant, to contribute to community-wide efforts to combat crime, and to rely on and cooperate with the police.

This 'coproductive' approach to community policing holds the potential to address the risks and pitfalls of more traditional models of community policing in several important ways. First, by harnessing the collective action potential of communities to serve as a "force multiplier" for the police, the coproduction approach to community policing may help address the manpower constraints that so often prevent police from responding effectively to crime, and thereby may help reduce the risk that greater reliance on police alone will prove either ineffective or counterproductive in combatting crime. Second, coproduction stands to be a more effective way to manage expectations in the process of building trust and confidence: rather than focus exclusively on improving citizens' expectations of police competence and trustworthiness, only to risk being unable to meet them, the coproduction model requires that police openly acknowledge their constraints in their bid to elicit coproduction from citizens. And finally, the coproduction model recognizes that communities living in the shadow of the state will invariably seek to self-provide protection, and that, in the absence of lawful alternatives, the risk that these efforts will be directed towards practices that *substitute* for the state, such as vigilantism and mob violence, is high. Whereas traditional models largely ignore local forms of security, and thus do little to remedy their more problematic aspects, the coproduction model seeks to address extra-legal practices through diversion — i.e. by directing communities towards lawful, rights-respecting activities which complement, rather than substitute for, the police.

The coproduction model of community policing may help address the risks and pitfalls of traditional community policing programs, but it is not without risks of its own. Delegating security is inherently risky, and could potentially increase rights abuses if watch groups do not abide by established rules and guidelines. Alternatively, the level of collective action from communities required to sustain these groups may prove difficult to achieve in practice. Communities may view watch groups as nonviable and therefore not worth their time and effort, or they may lack the collective action capacity to sustain them overtime. Police, for their part, may lack the organizational capacity to mobilize, track, and manage these groups; they may find them to be difficult to manage or 'illegible', with constantly shifting memberships and unclear leadership structures; or they may view them as a threat to their authority (Tapscott, 2017). In light of these barriers, the coproduction approach to community policing may well fail to engender meaningful levels of coproduction or collective action from citizens.

Although coproductive models of community policing have become increasingly common in fragile states — and increasingly popular among the donors that support them — there remains little rigorous evidence to inform whether the potential benefits of this approach indeed outweigh the risks.

3 Setting

Liberia is a small West African nation of approximately 4 million people. Between 1989 and 2003, the country experienced two devastating civil wars that killed over two hundred thousand civilians, displaced a large majority of the population, and left the country's government in a state of collapse. Efforts to reform the security sector began in 2004 under the direction of the UN peacekeeping mission (UNMIL). Since then, the government and its international partners have invested hundreds of millions of dollars to train and equip Liberia's police force. Yet after more than a decade and a half of intensive reform, Liberia's police force still lacks effective presence in most neighborhoods and much of the country remains plagued by high levels of crime and insecurity. This is especially true in Monrovia, Liberia's capital city and home to roughly 70% of the country's residents, many of whom live in densely-populated informal settlements. According to Afrobarometer data collected in 2016, 65 percent of the city's residents reported that they or someone they knew was a victim of theft in the past year, 35 percent reported that they or someone they knew was physically assaulted, and 78 percent said they felt unsafe in their neighborhood "several times", "many times", or "always" in the past year. Across all three sets of questions, these figures are the highest of any of the 36 capital cities covered by Afrobarometer's Round 6 Survey.

While there are many factors that contribute to Monrovia's high crime rate, capacity constraints within the Liberian National Police (LNP) are widely seen as playing a central role. In 2009, the International Crisis Group described the LNP as "an institution that has serious management deficiencies, few working vehicles and scant communications equipment, often lacks even handcuffs or torchlights and still suffers from a widespread perception of malpractice" (International Crisis Group, 2009, 17). More than a decade later, this remains an accurate description of the force.

Among citizens, the police are widely perceived as "insufficiently motivated to adequately respond to crime," and "lacking the resources to patrol or be proactive in crime prevention" (Reeve and Speare, 2012, 8). This lack of confidence has made many citizens reluctant to cooperate with the police through activities like crime reporting, information sharing, and evidence provision, further hindering their ability to combat crime. According to survey data collected in 2012, fewer than half of crimes that occur are ever reported to the police (Afrobarometer 2012). Among those that are reported, attending officers often complain of difficulty persuading citizens to come forward with information or evidence. As a result, only a small fraction of reported cases are ever solved or prosecuted (Human Rights Watch, 2013).

In addition to lacking confidence in the capacity of the police, many residents are unfamiliar with the law, the criminal justice system, and the procedures and costs associated with reporting crimes to the police. For victims of crime, this lack of awareness means they must spend considerable time and effort to learn about police procedures before reporting crimes, potentially deterring them from reporting at all.

Community policing in urban Liberia

Exasperated by a lack of security and frustrated by poor police performance, many communities have elected to self-provide security by organizing local, community-based security groups. The origins of these groups date to the period of crime and lawlessness that followed the end of the civil war in 2003, when communities formed vigilante groups to protect themselves from criminal gangs and armed robberies. Supported by donations of food, tea, and sometimes money from community members, these groups are often implicated in acts of mob violence.

Recognizing the need to build trust, educate citizens about the criminal justice system, reign in local security groups with no police affiliation, and provide an alternative to vigilantism, the LNP with support from its international partners initiated a significant expansion of its community policing program in 2014. In Monrovia, two activities have been central this expansion. First, in each of Monrovia's ten administrative police zones, the LNP created outreach offices staffed by Community Policing Officers (CPOs) with special training in community outreach. In addition to their regular duties as patrol officers, the CPOs are responsible for organizing town hall meetings in communities on a semi-regular basis. During these meetings, officers educate residents about the criminal justice system, solicit information about security threats and brainstorm strategies to address them, and provide residents with the opportunity to ask questions or express concerns. Alongside these meetings, officers conduct foot patrols in which they interact with residents in small groups, solicit additional information about security concerns, and distribute informational pamphlets that reinforce the content communicated during the town hall meetings.

In the second component of the program, the CPOs use the town hall meetings as an opportunity to (re)introduce community leaders and residents to the Watch Forum initiative. The CPOs explain that Watch Forums are composed of groups of concerned citizens who assist the police by sharing information about security threats; meeting regularly with the police to design proactive, collaborative strategies to combat crime; facilitating police investigations in their communities; and conducting nighttime security patrols during periods of peak crime.

The LNP's support for this two-pronged approach to community policing is founded on the hope that town hall meetings, foot patrols and educational pamphlets will provide residents with the knowledge, familiarity, and confidence they need to rely on the police, while encouraging communities to form Watch Forums will help to direct them towards lawful, coproductive forms of security provision that strengthen rather than undermine state authority and help address police capacity constraints. This model remains untested, however, and comes with several potential limitations and risks, as discussed in Section 2.

4 Research Design

4.1 Sampling & Randomization

Monrovia is divided into ten administrative police zones, which are akin to police precincts in major U.S. cities and typically composed of between 15 and 40 communities or neighborhoods. Within each zone, local research assistants worked with the CPOs to identify any 'high priority' communities to be nominated for the intervention based on their assessments of high crime rates and/or strained police-community relations. This process identified 35 'high priority' communities. Because this sample size was smaller than anticipated and would have resulted in an underpowered study, an additional 65 communities were randomly sampled from the remaining population of communities to yield a target sample size of 100 communities. Half of the communities within each zone were then randomly assigned to treatment via block (i.e. zone) randomization. During the baseline survey, two communities were found to be duplicates of other communities and were dropped; during implementation, staffing constraints within the implementation monitoring team required that the smallest police zone be dropped. These two changes resulted in a final sample size of 93 communities, 45 of which were assigned to treatment. Appendix A.5 shows that treatment and control communities are well-balanced on baseline outcome indices and covariates:

differences for only two out of twenty variables tested have a $p - value < 0.05$.

Further details on the sampling procedure for this study, including a discussion of minimum distances between communities that minimize the risk of spillover, are discussed in Appendix A.3.

4.2 Intervention & implementation

Implementation began in February 2018 and continued for a period of ten months. In each of the 45 treatment communities, town hall meetings were held approximately every other month, usually on weekend afternoons, and were preceded by foot patrols during the week in which officers distributed informational pamphlets and alerted residents to the upcoming meeting. In total, each community hosted between 5 and 6 meetings and accompanying foot patrols. Attendance ranged from as little as 25 residents to as many as 80, but on average was 44 adult residents. Implementation of the program was monitored by research assistants from a local NGO who worked in close collaboration with the CPOs and accompanied them on all meetings and foot patrols, taking detailed notes on the proceedings, the topics covered during the meetings, and the questions raised during the Q&A.

The meetings covered a variety of topics, including: the Watch Forum initiative; the 'concept' of community policing and the importance of police/community partnerships; the procedures for reporting crimes to the police; the Professional Standards Division of the LNP and its role in handling incidents of police misconduct; the Women and Children Protection Services division of the LNP and its role in handling domestic disputes and child endangerment; and the names and phone numbers of 'key contacts' at the local police station. Major themes from the research assistants' Observation Notes, along with the pamphlets used to summarize the messages communicated during the meetings, can be found in Appendix A.1.

4.3 Hypotheses

I pre-specified 16 hypotheses, 14 of which I test here.[2] My hypotheses centered on two broad pathways through which I expected the program to reduce crime and insecurity. First, I considered the program's potential impact on citizens' expectations of the costs and benefits of cooperation, and how changes in cooperation would in turn influence police effectiveness and crime. Second, I considered the program's potential impact on community coordination and collective action in coproductive forms of self-protection, and how these changes would in turn influence police effectiveness and crime. I organize these hypotheses into three mechanism categories — costs, benefits, and coproduction — as depicted in Table 1.

These three sets of mechanism hypotheses underlie the following primary hypotheses. First, as a result of lower costs and greater benefits to cooperation, I expected individual-level forms of cooperation to increase, as measured by information sharing (H1), crime reporting (H2), and willingness to report acts of police abuse (H3). Due to these changes as well as greater community coproduction, I expected the police to become more effective, leading to lower crime (H4), improved perceptions of security (H5), and greater satisfaction with police performance (H6).

In addition to these pre-specified hypotheses, this study tests whether the program reduced the incidence of mob violence (H7) and whether it improved knowledge of the rules governing community watch groups / forums (M3b). Although not pre-registered, these hypothesis are a direct corollary of my pre-specified hypotheses on coproduction and attitudes towards mob violence. In addition, construction of the M3b index follows the same procedure as all the other pre-specified index variables, and includes every variable relevant to knowledge of the rules governing watch groups, which was a standalone section in the survey. Similarly, the incidence of mob violence (H7) is measured using the single survey variable on this topic. And both M3b and H7 are tested using the same pre-specified estimation equation as the other indices. Taken together, these con-

[2] My pre-analysis plan is available at: [REDACTED FOR PEER REVIEW]. Apart from the coproduction hypotheses, the hypotheses tested is this study were harmonized with the Metaketa initiative (Blair et al., 2021). The two that are excluded from this paper are not directly relevant to the topics covered here. Concretely, they pertain to secondary impacts on attitudes toward government, as well as officer-level changes of behavior, neither of which were expected in this study. Results for these hypotheses are reported in Appendix A.12.

siderations leave little to no room for false positive results due to p-hacking.

4.4 Monitoring and minimizing risks to human subjects

This study was conducted with IRB approval and adheres to the American Political Science Association's Principles and Guidance for Human Subjects Research. Going beyond these basic requirements, the research team took extensive efforts to monitor and minimize risks to human subjects in this study. These topics are discussed in detail in Appendix A.2.

5 Empirical Analysis

The empirical analysis in this paper follows my pre-analysis plan, which described in complete detail the procedure for constructing outcome variables, the estimation equations, and the procedure used to account for multiple comparisons.

5.1 Data

This study draws on three sources of data. First, I draw on data from baseline and endline surveys administered to a random sample of 20 adults in July 2017 (eight months prior to the start of the intervention) and January 2019 (three months after the end of the intervention), respectively, in each of the 93 study communities. In addition to background information on demographics, the surveys measured outcomes pertaining to each of my 15 hypotheses, as detailed below. Second, I draw on the LNP's administrative data on crimes reported during the same 18 month period. However, because these data capture only a fraction of crimes that are actually reported to the police,and because they do not allow us to distinguish between crimes that occur and crimes that were reported, they are not used for testing the hypotheses above. Results for administrative crime data are reported in the Appendix.

I also draw on data from a survey of patrol officers and a review of police logbooks at each of Monrovia's 32 police stations designed to capture information on incidents of mob violence.

These data were collected in March 2019 for purposes of informing and validating the results from the large-N survey. Further details on these data are provided in Section 6.4.

5.2 Outcome variables

Table 1 summarizes the hypotheses tested in this study. Each hypothesis is linked to a cluster of between two and fourteen variables linked to individual survey questions. To mitigate the risk of false positives, I test each hypothesis using a single composite index constructed by i) standardizing each variable by its baseline mean and standard deviation, and then ii) taking the mean across the individual variables in the cluster for each respondent.Appendix A.13 provides complete details on the construction of the composite indices and the corresponding variables and survey questions; Appendix A.4 reports summary statistics for each constituent variable, organized by outcome cluster.

Table 1 organizes the hypotheses and corresponding outcomes into two categories — mechanism hypotheses and primary hypotheses. When testing the primary hypotheses, I use the procedure outlined in Benjamini and Hochberg (1995) to adjust my p-values and control the risk of false discovery to 5%. In Appendix A.10, I report results individually for all the variables within each outcome cluster, also using the procedure outlined in Benjamini and Hochberg (1995) to control the within-cluster false discovery rate to 5%.

5.3 Estimation

The estimand in this study is the average treatment effect of the intervention on the sample communities (i.e. the SATE). To estimate this, I use weighted least squares (OLS) regression of the outcome on a dummy variable indicating whether or not the community was assigned to the treatment group, with weights constructed as the product of i) the inverse of the probability of assignment to treatment/control, and ii) the inverse of the probability that an individual was selected for the endline survey. The former accounts for the fact that the probability that a community is assigned to treatment or control varies across randomization blocks (i.e. police zones); the latter accounts

Table 1: Hypotheses and Outcome Indices

Mechanisms	Index name	Hypothesis	Pre-specified?
Costs of cooperation			
Increase familiarity with the police	know_pol_idx	M1a	Y
Increase knowledge of the criminal justice system	know_idx	M1b	Y
Improve perceptions of police intentions	intentions_idx	M1c	Y
Reduce social sanctions for reporting	norm_idx	M1d	Y
Benefits of cooperation			
Improve perceptions of police capacity	police_capacity_idx	M2a	Y
Improve perceptions of police responsiveness	pol_responsiveness	M2b	Y
Community coordination & coproduction			
Reduce norms supporting mob violence	sup_mobviol_idx	M3a	Y
Improve knowledge of rules governing watch groups / forums	know_cwt_idx	M3b	N
Increase contributions to community coproduction	ca_sec_idx	M3c	Y
Primary Hypotheses	Index name	Hypothesis	
Cooperation with police			
Increase reporting of crimes to the police	crime_reporting_idx	H1	Y
Increase crime tips & information sharing	tips_idx	H2	Y
Increase willingness to report police abuse	police_abuse_idx	H3	Y
Security			
Reduce the incidence of crime	crime_victim_idx	H4	Y
Improve perceptions of security	future_security_idx	H5	Y
Improve satisfaction with police performance	satis_idx	H6	Y
Reduce the incidence of mob violence	cmob_num	H7	N

Notes: Each index is a composite index constructed from several individual variables. For full details on the construction of the indices, see Appendix A.13.

for the fact that individuals from relatively large communities have a lower likelihood of being included in the endline sample compared to those from relatively small communities.

Standard errors are clustered at the community level to account for the cluster-randomized design, and all specifications include block fixed effects. For outcomes measured at both baseline and endline, I also include the community-level average outcome at baseline as a control variable (because the baseline-endline survey is not an individual-level panel, I cannot control for baseline outcomes at the individual level).

5.4 Threats to inference

Measurement error

Most of the outcomes in this study are self-reported. This introduces the possibility of response bias correlated with treatment. For example, if respondents associate the endline survey with the intervention, this could cause them to underreport certain crimes or incidents, or to report attitudes consistent with messages conveyed during the program (e.g. opposition to mob violence), introducing bias. While I cannot rule out this possibility entirely, I can point out that, taken as a whole, the patterns reported below are inconsistent with social desireability bias: although some effects align with social desirability bias towards program messages, effects on several of the outcomes most closely associated with program messages and the risk of bias are null. In addition, in Section 6.4, I replicate the main results on mob violence using a completely independent data source that is not subject to the risk of social desirability bias correlated with treatment.

Spillover

Because communities are large and populous, and because no two communities in the sample are adjacent, the risk of spillover is likely to be low. I discuss this risk in detail in Appendix A.3.

6 RESULTS

6.1 INTENSITY OF TREATMENT

From administrative implementation data, we know that treatment communities on average received 5.4 town hall meetings, with most meetings attended by between 20 and 30 people. Table 2 tests whether these activities resulted in a measurable increase in police presence in the eyes of citizens. The results indicate that community interactions with police were .67 standard deviations higher in treatment communities than in control communities, an effect that's driven by greater exposure to town hall meetings with police rather than by more frequent foot or vehicle patrols. Substantively, the proportion of residents reporting they had attended a meeting with the police in the past six months increased by 30 percentage points in treatment communities, to 40 percent from a mean of 10 percent in control communities.

Table 2: Intensity of treatment

	(1) Intensity of treatment index (std)	(2) Sees foot patrols daily or weekly	(3) Sees vehicle patrols daily or weekly	(4) Attended police mtg past 6m
Treatment	0.67^{***}	0.01	0.02	0.30^{***}
	(0.06)	(0.02)	(0.02)	(0.03)
Covariates	Y	Y	Y	Y
Ctrl_mean	0.06	0.11	0.12	0.10
Observations	1836	1851	1851	1851

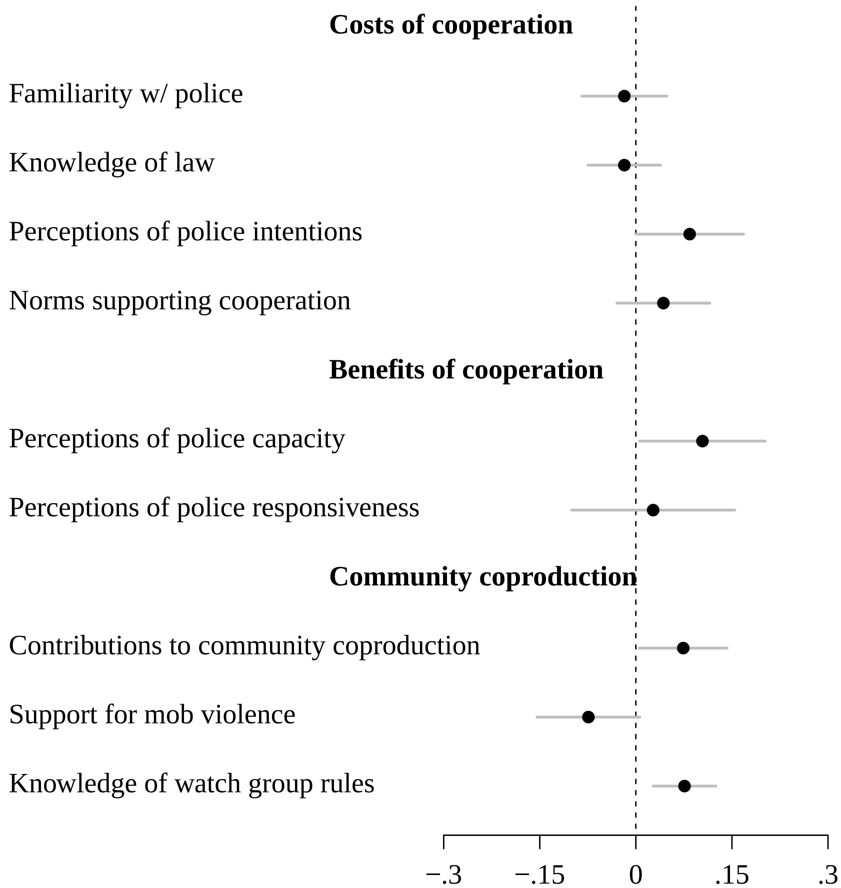
Robust standard errors in parenthesis, clustered by community. Full results in Appendix A.6. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$

These results suggest that the intervention's impacts on the primary and secondary outcomes in this study were driven primarily by changes precipitated by the town-hall meetings rather than by changes linked to police presence, such as deterrence during foot patrols or problem-oriented policing. They also suggest that whatever follow-up actions the police took in response to information acquired through the meetings did not involve a significant increase in police presence.

6.2 Impacts on hypothesized mechanisms

Figure 1 plots average treatment effects on composite indices for each of my hypothesized mechanisms. The program did not improve residents' overall familiarity with the police, as measured by the composite measure constructed from questions about the knowledge of police services (e.g. the existence of a women's and child protection unit) and personal familiarity with local police officers (e.g. knowing the name and phone number of an officer). It also did not improve knowledge of Liberia law, as measured by questions on habeas corpus, the right to legal counsel, the legality of "case registration fees," and laws on statutory rape and child abuse, among other basic legal questions.

Figure 1: Effects on Hypothesized Mechanisms



Notes: Average Treatment Effects on indices for hypothesized mechanisms. All indices standardized by baseline mean and standard deviation. $N = 1851$. See Appendix A.7 for results in regression table format.

Figure 1 also suggests the program did not reduce the social costs that citizens face when reporting crimes. Substantively, residents of treatment communities were about as likely to say that community members become angry when you report burglaries, land disputes, or domestic violence to the police, as compared to residents of control communities. They were also comparably likely to say that providing the police with information to solve crimes can lead to backlash.

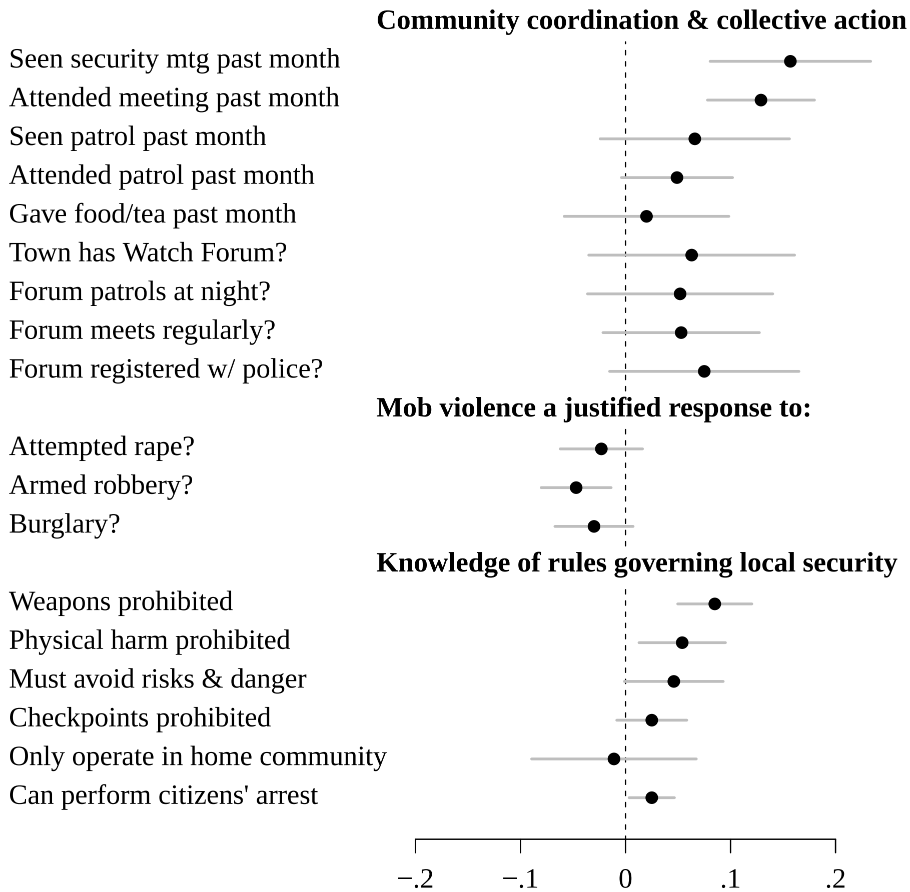
Despite these null effects, the program was successful at improving perceptions of police intentions and capacity. Most notably, residents of treatment communities were six percentage points less likely to view the police as corrupt (a decrease of 12% relative to the control group mean); five percentage points more likely to say the police treat all citizens fairly (12%), six percentage points more likely to say the police care about the well-being of residents (9%), and seven percentage points (12%) more likely view the police as capable of responding to crimes in a timely manner. Changes in perceptions on other dimensions of intentions and capacity were modest and not statistically significant, but always positive (Appendix A.10).

The program's largest and most consistent impacts were on the set of outcomes targeted by the Community Watch Forum initiative. Here, I find large and substantively meaningful effects on community collective action and contributions to local security groups. As reported in Figure 2, residents of treatment communities were 13 percentage points more likely to attend a security meeting in the past month, four percentage points more likely to participate in a security patrol, six percentage points more likely to report that a community watch forum exists in their community, six percentage points more likely to report the watch forum meets at least monthly, and seven percentage points more likely to report the watch forum was registered or affiliated with the police, as compared to residents of control communities.[3] As a whole, these improvements amount to a .18 standard deviation increase in the community coordination and coproduction index. It is important to note that because the one-month recall period for these actions does not overlap with the intervention, which ended three to four months prior to the endline, these effects are not merely a function of participation in the intervention itself.

Importantly, greater coproduction by communities was accompanied by greater knowledge of the rules governing local security groups. As compared to residents of control communities, residents of treatment communities were: nine percentage points more likely to know security groups are not permitted to carry weapons, five percentage points more likely to know they cannot beat an uncooperative suspect into submission, four percentage points more likely to know they should not enter into dangerous situations, and four percentage points more likely to know they

[3] There is considerable variation across police zones in what it means for a Watch Forum to be "registered" with the police. The survey question on registration was not written on the assumption that respondents would necessarily be well-informed about the registration process or their Forum's registration status; rather it was meant to capture whether residents perceive the Forum to be working in coordination with the police (e.g. because they see police attend Forum meetings, or because they see group members walking or talking with police in their community).

Figure 2: Effects on coproduction disaggregated by component variables for indices



Notes: Average Treatment Effects on component variables for coproduction indices. All outcome variables are binary. $N = 1851$. See Appendix A.8 for results in regression table format.

can detain a suspect until police arrive provided they do not cause harm.

And finally, the program appears to have reduced support for mob violence, as measured by whether respondents believe mob violence would be "justified" or "somewhat justified" in response to three hypothetical scenarios of crime. While this effect is only marginally significant ($p < 0.1$), it is consistent with the findings reported above and below, lending credence to the assumption that it is not a Type I error.

6.3 Impacts on primary outcomes - cooperation, crime, and security

The LNP's community policing program was designed to reduce the costs of cooperation, increase the expected benefits, and catalyze coproduction from citizens. These changes, in turn, were expected to increase cooperation, reduce reliance on extralegal practices, and improve police effectiveness, potentially leading lower crime and greater security.

Thus far, the results suggest the program may have increased the expected the benefits of cooperation and reduced the expected costs by improving perceptions of police capacity and intentions, but that overall changes to individuals' cost-benefit calculations were likely modest due to the null effects on knowledge of Liberian law, familiarity with police services, and perceptions of police responsiveness. Improvements in coproduction, by contrast, were large and substantively meaningful, and could potentially have contributed to greater security either directly (e.g. by deterring crimes), indirectly (e.g. by facilitating police work), or by substituting legal forms of coproduction for violent, illegal practices.

This section tests for such downstream impacts. As above, I report average effects on the composite index for each hypothesis in the main text of the paper and effects on the individual variables that make up each index in the appendix.[4] The results, reported in Figure 3, indicate the program did not improve individual forms of cooperation with the police, as measured by i) willingness to report crimes to the police, ii) provision of information or crime tips to police in the past six months, or iii) willingness to report police misconduct. The program also did not significantly increase the likelihood that victims of crime subsequently reported their case to the police, as I show in the Appendix A.12).

As alluded to above, these null effects may reflect the fact that changes to individuals' assessments of the costs and benefits of cooperation were too modest to influence cooperation or reliance. Indeed, during the town hall meetings, instances of police failing to follow-up on crime tips or ad-

[4]Results on the indices for each of my primary hypotheses with multiple comparisons adjustments are also reported in the Appendix. Levels of statistical significance are unchanged from those reported here.

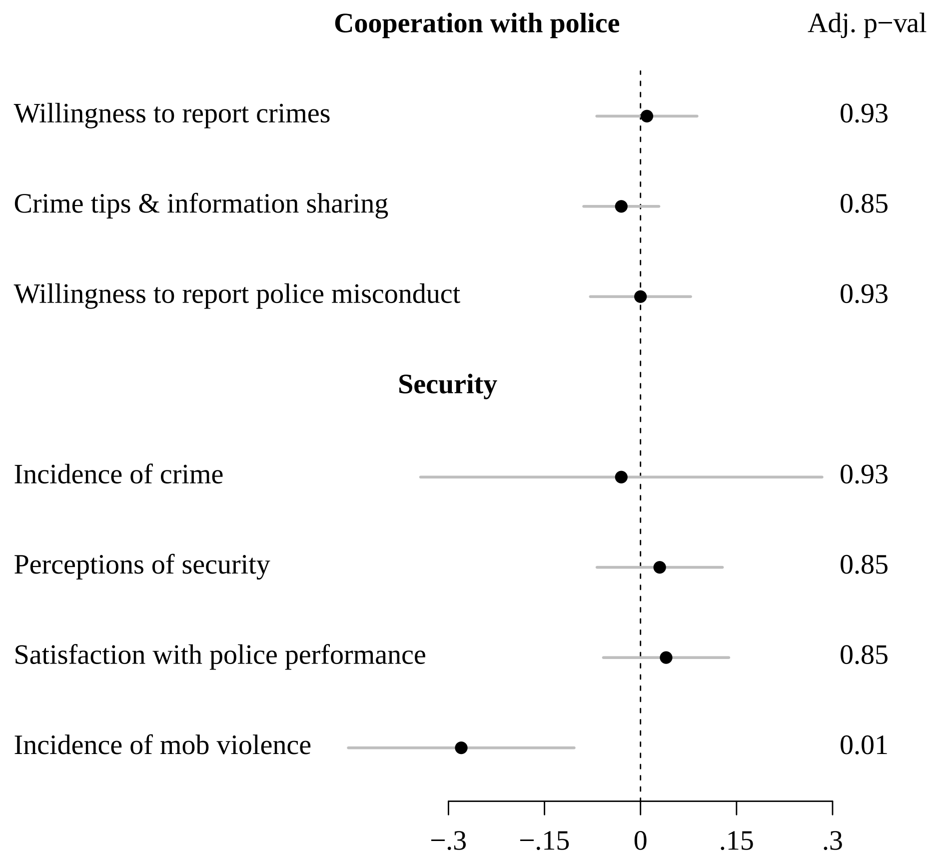
equately investigate crimes came up repeatedly, complaints which officers openly acknowledged as issues they struggle with. Given that the intervention did not materially change police capacity or resources, residents' grim assessments of the value of reliance may have (justifiably) remained unchanged.

The program also did not meaningfully reduce the overall incidence of crime or most of the particular categories of crime measured in the survey, including: armed robbery, burglary, aggravated assault, simple assault, sexual violence, domestic violence, and violent and non-violent land disputes. In the appendix, I replicate the null overall effect on crime using the LNP's administrative data on reported crimes, lending additional weight to the conclusion that the program did not reduce crime.

Perhaps unsurprisingly given the null effects on crime, the program appears to have had little overall impact on perceptions of security. Substantively, residents of treatment communities were about as likely to fear violent or non-violent crime in the future or to have feared crime or felt unsafe walking in their community in the past six months, as compared to residents of control communities. They were, however, four and five percentage points, respectively, more likely to say they would feel safe leaving their generator or motorbike outside their home at night — modest though statistically significant effects that could potentially reflect greater nighttime activity by local security groups.

Perhaps the most important finding reported in Figure 3 is the .16 standard deviation reduction in the incidence of mob violence. Substantively, residents of treatment communities reported an average of .34 fewer instances of mob violence in their communities in the past year, a reduction of 39% relative to the control group mean.Notably, these results do not appear to be a function of social desirability bias. Differences between treatment and control communities in the incidence of other types of crime that could potentially been seen as socially undesirable by respondents, such as riots, murders or any of the categories of crime measured in the crime index, are small and statistically insignificant. Moreover, we do not observe positive effects on other sets of outcomes potentially seen as socially desirable, such as willingness to report crimes or the provision

Figure 3: Effects on Primary Outcomes



Notes: Average Treatment Effects on indices for primary outcomes. All indices standardized by baseline mean and standard deviation. P-values adjusted following (Benjamini and Hochberg, 1995). $N = 1851$. See Appendix A.9 for results in regression table format.

of information to police in the past 6 months, as reported above.

6.4 Robustness of effects on mob violence

It is not possible to validate the mob violence finding using administrative crime reports from the LNP because incidents of mob violence are seldom recorded in official crime reports, even when reported to police. Indeed, of the 17,508 incidents included the LNP's database of crimes occurring between July 2017 and January 2019, there is only one report of mob violence, despite the fact that there were at least a dozen high-profile incidents during this timeframe that *did* involve a police response and were reported in the local press or on social media.

In light of the limitations to administrative data, I assigned a team of research assistants to visit each of Monrovia's 32 police stations in March 2019 — three weeks after the endline concluded and two weeks after my initial analysis — to collect independent information on incidents of mob violence from patrol officers. As street-level agents with a close working familiarity of communities, officers are well-positioned to serve as key-informants for incidents of mob violence, including those that never made it into police records. And because they did not participate in or know about the intervention, they are also not subject to the risk of social desirability bias correlated with treatment.[5]

The research assistants first reviewed the police station logbooks and met with on-duty patrol officers to inquire about any known incidents of mob violence in the past seven months (from August 2018 to February 2019). For any identified incident, an officer with first or reliable second-hand knowledge of incident was administered a short survey about the incident. The survey covered the approximate date and time of the incident, the events leading up to the incident, the outcome of the incident, and the location of the incident.[6]

The resulting dataset includes 41 incidents. All but two were precipitated by an act of petty theft, such as stealing a phone or handbag in a market, and about half occurred at or after dusk. None of the incidents resulted in death, but 73% (30/41) required treatment at a hospital. None of these incidents were recorded in the LNP's electronic database, and only 20% (8/41) were documented in police logbooks.

Of the 41 incidents documented by the key informant survey, 15 occurred in communities included in this study's sample – 11 in control communities and 4 in treatment communities. Table 3 reports these results in a community-level OLS regression framework, following the pre-specified

[5]None of the officer informants overlapped with those participating in the intervention. In addition, because the intervention was organized from the headquarters station for each of the nine zones, officers from only nine of the 32 stations would have had colleagues that participated in the intervention. Lastly, even officers who conducted the intervention were not actually aware of the evaluation study — they were simply told by the CPO and their commanding officer when and where they would be participating.

[6]In some cases, on-duty patrol officers reported that one of their off-duty colleagues was most knowledgeable about the incident, in which case the survey was scheduled for their next daytime shift.

Table 3: Effect on mob violence (police survey)

	Acts of mob violence in community, past 8 months	
Treatment	-0.14^{+}	-0.15^{+}
	(0.08)	(0.08)
Ctrl group mean	0.23	0.23
Block fixed effects	N	Y
N	93	93

Notes: Community-level regression of the # of mob violence incidents on treatment. Standard errors in parentheses. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

estimation equation for community-level outcomes documented in my pre-analysis plan. The results show that treatment reduced the likelihood of mob violence occurring in treatment communities by 14 percentage points ($p < 0.07$), or about 60% relative to the control group mean. The effect is only marginally significant, but it is consistent with the findings from the citizen survey in its direction and magnitude relative to the control group mean.

7 Discussion

7.1 Mechanisms underlying reductions in mob violence

What explanations might account for the intervention's negative effect on mob violence? While it is impossible to answer this question definitively, two potential explanations stand out as plausible.[7] First, the intervention may have altered the knowledge, attitudes, and norms of members of local security groups, making them more supportive of the rule of law and more integrated into formal policing practices. As a corollary, the intervention may have removed members with a reputation for violence or criminal behavior (e.g. through the vetting process associated with the Watch Forum initiative).As a result of these changes, local security groups may have become less likely to perpetrate acts of vigilantism and more likely intervene to stop them from occurring.

[7] Note that it is unlikely that the intervention had any deterrent effect on vigilantism, as police presence did not increase outside the town hall meetings.

A second potential explanation is that the intervention altered the knowledge, attitudes and norms among residents, making them more aware of how they can respond to crimes in a lawful manner by supporting or calling upon Watch Forums, more supportive of the rule of law, and less likely to resort to vigilantism.

One way to evaluate the plausibility of the first explanation is to test whether those who participate in the provision of local security differ across treatment and control communities in terms of their norms, attitudes, and orientation towards the police. To this end, I regress indices for the attitudinal and normative outcomes in this study — familiarity with police, knowledge of the law, norms against cooperation, perceptions of police capacity, intentions, and responsiveness, support for mob violence, and knowledge of Watch Forum rules — on an indicator for treatment while subsetting to respondents who report either attending a security meeting in the past month or participating in a security patrol.[8] Because participation is post-treatment, the results of this analysis do not have a causal interpretation, but they can be interpreted descriptively as an indication of whether there are statistically significant differences between local security providers in treatment vs. control communities.

The results, reported in Table 4, indicate that local security providers in treatment communities are more optimistic about police capacity and intentions, more knowledgeable of the rules governing their conduct, (weakly) less supportive of mob violence and (weakly) more supportive of cooperation with the police. Further support for the idea that the intervention made local security providers more supportive of the rule of law is found in the Observation Notes recorded by local research assistants during the town hall meetings, which document several instances in which members of Community Watch Forums acted to diffuse situations that otherwise could have escalated into mob violence (Appendix A.1).

The results reported in this paper are also consistent with the second explanation, whereby the intervention's impact on knowledge, norms, and attitudes among residents as a whole made them less likely to resort to vigilantism. Not only were residents of treatment communities less

[8] 26% of respondents from control communities attended a meeting or participated in a security patrol in the past month, as compared to 37% in treatment communities.

Table 4: Differences between members of local security groups in treatment vs. control communities

	Familiarity w/ police	Knowledge of law	Police intentions_idx	Norms of cooperation
Treatment	-0.03	-0.02	0.14^{*}	0.09^{+}
	(0.05)	(0.03)	(0.06)	(0.05)
N	581	581	526	564
	Police Capacity	**Police Responsive**	**Support for mob violence**	**Knowledge of Watch Forum Rules**
Treatment	0.20^{**}	0.14	-0.11	0.08^{*}
	(0.07)	(0.10)	(0.07)	(0.04)
N	575	571	573	581

Notes: Clustered standard errors in parentheses.
$^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

supportive of mob violence, they were also more aware of the presence of local groups operating in coordination with the police in their communities. In particular and as previously reported in Figure 2, they were considerably more likely to either participate in security meetings or group activities (30%) or to be aware of them happening (50%), relative to residents of control communities (at 18% and 35%, respectively). In addition, analysis of additional outcomes from the endline survey indicates they were: eight percentage points more likely to agree or strongly agree with the statement “My community is well-organized in the fight against crime” ($p < 0.18$); thirteen percentage points more likely to agree or strongly agree that “Leaders in my community have a close working relationship with the police”($p < 0.03$); nine percentage points more likely to agree or strongly agree that “Members of my community do a good job of working together to help the police” ($p < 0.08$); and five percentage points more agree or strongly agree that “Leaders in my community do a good job of organizing my community to combat crime” ($p < 0.38$). While these differences do not constitute direct evidence of substitution away from vigilantism, they do suggest that residents of treatment communities became more aware of Watch Forums and their ability to respond to crimes in a lawful manner that complements rather than substitutes for the police, and that their perceptions of the efficacy and accessibility of these groups improved.

7.2 Coproduction and the risk of adverse impact

Coproduction plays a central role in helping resource-constrained governments provide essential public services (Post, Bronsoler and Salman, 2017). However, because it requires the state to delegate authority and concede direct control of service provision, coproduction can also compromise accountability and equity in access to services (Cammett and MacLean, 2014). This is especially true of coproduction in the security sector, where the power afforded to non-state actors has the potential to undermine informal sources of accountability and enable predatory behavior (Baranyi, Beaudet and Locher, 2011, 135). In extreme cases, collaboration with non-state security actors can give rise to vigilante groups or militias that become emboldened to acquire political power and resist states' efforts to reassert control (Hidalgo and Lessing, 2015; Krasner and Risse, 2014).

In the case of Liberia's model of community policing, these risks do not appear to have materialized. On the contrary and as presented in Table 4 above, security providers in treatment communities became more supportive of the rule of law and more inclined to cooperate with police. As an additional test for potential adverse effects of coproduction models of community policing, I assess its impact on perceptions of community leaders, who were actively involved in facilitating the intervention and coordinating security groups, on the assumption that perceptions of leaders would decline if local security groups lacked accountability or became involved in predatory behavior.

Table 5 reports the results of this analysis, following the same estimation procedure used for the main analysis and outlined in Section 5.3. Contrary to the idea that mobilizing communities to form local security groups empowers 'local despots', the results suggest the intervention actually *improved* perceptions of local leaders,[9] who played a central role in organizing the intervention and managing the Watch Forums. Substantively, residents of treatment communities were about 6 percent more likely to believe leaders treat all residents the same, 4 percent less likely to view leaders as corrupt, and 4 percent more likely to believe their leaders rule transparently, relative to

[9] In Monrovia, it is customary for each community to have a town chairperson, a leader for each neighborhood block, and various miscellaneous leaders such as the the women's group and youth group leaders. These persons are not government employees and are typically elected through informal community elections.

residents in control communities. Not all of these differences are significant, but they all point in the same direction, and the effect on the composite index (Column 1) *is* statistically significant.

Table 5: Effects on perceptions of local leaders

	Perceptions index (std)	Leaders treat all equal?	Leaders corrupt?	Leaders rule transparently?
treatment	0.14*	0.13+	-0.09	0.09
	(0.06)	(0.07)	(0.06)	(0.06)
Ctrl mean	-0.08	2.03	1.62	2.37
N	1851	1851	1851	1851

Robust standard errors in parentheses. Outcomes in columns 2-4 measured on a four-point, agree-disagree scale. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$

8 Conclusion

Police in weak states face many obstacles in trying to provide security, not least of which is a severe shortage of capacity. In the absence of effective policing, communities often resort to extralegal practices such as mob violence, lynching, and vigilantism. Community policing programs have been at the forefront of governments' efforts to address this challenge. These programs aim to do this by building confidence in the police, dissuading citizens from reliance on extralegal practices, and providing legal, rights-respecting pathways through which communities may coproduce security.

But there is little evidence on whether these programs work as intended. I address this gap by experimentally evaluating the Liberian National Police's coproduction model of community policing. I show that program improved familiarity and perceptions of police and mobilized communities to coproduce security by forming local security groups and incorporating them into the police's Watch Forum initiative. I further show that these changes were accompanied by a roughly 40 percent reduction in the incidence of mob violence.

I also find, however, that despite improvements in trust and coproduction, the program did not reduce crime, improve perceptions of security or increase crime reporting. I argue that these null effects most likely reflect the fact that capacity constraints within the LNP remained unchanged by the intervention. As a result, efforts by communities and security groups to provide crime tips or facilitate police investigations were seldom met with an effective police response, and most residents remained skeptical of the value of reporting crimes or otherwise cooperating with police.

These findings have important implications for the literature on coproduction as a strategy to supplement weak state capacity in developing countries. Much of this literature assumes that coproductive strategies are most important precisely where capacity constraints are most severe, because states with low capacity can least afford to provide services on their own. My findings, on the other hand, suggest a certain baseline level of capacity may be a prerequisite for successful coproduction. When capacity is so low that government institutions cannot reciprocate citizens' contributions with contributions of their own, coproduction is unlikely to improve the quality of services.

My findings also suggest, however, that while coproduction cannot substitute for the shortcomings of weak states, it can be effective at drawing communities away from harmful or rights-abusing practices. This finding has important implications for other service sectors where provision by non-state actors may conflict with international standards or human rights, such as the justice, health, and education sectors. When designing policies in these and other sectors where non-state providers are prevalent, policymakers should consider coproduction not only for its potential to supplement state capacity, but also for its potential to divert non-state actors away from harmful practices.

Lastly, this study contributes to the broader literature on state-building, which has long equated states' success at centralizing and consolidating power with their ability to engender compliance with the law (Migdal and Schlichte, 2005). Contrary to this perspective, I show that *delegating* authority to local leaders and communities through coproduction can actually *strengthen* compliance with the law by converting potential rule-breakers into law-abiding coproducers. This finding is consistent with prior work on coproduction in more developed states, which shows that by bringing citizens and state officials together for a common cause, coproduction can build trust and social capital that spills-overs into increased compliance (Tsai, 2011; Cammett and MacLean, 2014). I extend this literature by showing that coproduction can lead to a similar dynamic in the security sector in weak states, improving relations between police and citizens, engendering norms against breaking the law, and integrating otherwise peripheral communities into the state's process of providing public goods. While further research is needed to verify the scope of this potential, coproduction appears to be a promising strategy for building trust and compliance in regions where the state and its rule of law are weak.

References

Abrahams, Raphael Garvin. 1998. *Vigilant Citizens: vigilantism and the state*. Polity Press.

Andrews, Matt, Michael Woolcock and Lant Pritchett. 2017. *Building state capability: Evidence, analysis, action*. Oxford University Press.

Baker, Bruce. 2009. *Security in post-conflict Africa: the role of nonstate policing*. CRC Press.

Baker, Bruce and Eric Scheye. 2007. "Multi-layered justice and security delivery in post-conflict and fragile states: Analysis." *Conflict, Security & Development* 7(4):503–528.

Baranyi, Stephen, Pierre Beaudet and Uli Locher. 2011. "World development report 2011: conflict, security, and development.".

Benjamini, Yoav and Yosef Hochberg. 1995. "Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing." *Journal of the Royal Statistical Society. Series B (Methodological)* pp. 289–300.

Blair, Graeme, Jeremy M Weinstein, Fotini Christia, Eric Arias, Emile Badran, Robert A Blair, Ali Cheema, Ahsan Farooqui, Thiemo Fetzer, Guy Grossman et al. 2021. "Community policing does not build citizen trust in police or reduce crime in the Global South." *Science* 374(6571):eabd3446.

Blair, Robert, Sabrina Karim and Benjamin Morse. 2019. "Establishing the rule of law in weak and war-torn states: Evidence from a field experiment with the Liberian National Police." *American Political Science Review* .

Boege, Volker, M Anne Brown and Kevin P Clements. 2009. "Hybrid political orders, not fragile states." *Peace Review* 21(1):13–21.

Brunson, Rod K. 2007. "Police don't like black people?: African-American young men's accumulated police experiences." *Criminology & Public Policy* 6(1):71–101.

Cammett, Melani and Lauren M MacLean. 2014. *The politics of non-state social welfare*. Cornell University Press.

Carothers, Thomas. 2006. *Promoting the rule of law abroad: in search of knowledge*. Brookings Institution Press.

Chirayath, Leila, Caroline Sage and Michael Woolcock. 2005. "Customary law and policy reform: Engaging with the plurality of justice systems.".

Darley, John M, Kevin M Carlsmith and Paul H Robinson. 2000. "Incapacitation and just deserts as motives for punishment." *Law and Human behavior* 24(6):659–683.

Denney, Lisa. 2014. *Justice and Security Reform: development agencies and informal institutions in Sierra Leone*. Routledge.

Denney, Lisa and Sarah Jenkins. 2013. "Securing communities: The what and the how of community policing." *Overseas Development Institute Background Paper* .

Dinnen, Sinclair and Gordon Peake. 2013. "More than just policing: Police reform in post-conflict Bougainville." *International peacekeeping* 20(5):570–584.

Downie, Richard. 2013. *Building police institutions in fragile states: case studies from Africa*. Center for Strategic and International Studies.

Ferreira, Bertus R. 1996. "The use and effectiveness of community policing in a democracy." *Policing in Central and Eastern Europe: Comparing Firsthand Knowledge with Experience from the West. College of Police and Security Studies, Ljubljana, Slovenia* .

García-Ponce, Omar, Lauren Young and Thomas Zeitzoff. 2018. "Anger and Support for Punitive Justice in Mexico?s Drug War.".

Gerber, Alan S. and Donald P. Green. 2012. *Field Experiments: Design, Analysis, and Interpretation*. New York: W. W. Norton.

Gill, Charlotte, David Weisburd, Cody W Telep, Zoe Vitter and Trevor Bennett. 2014. "Community-oriented policing to reduce crime, disorder and fear and increase satisfaction and legitimacy among citizens: A systematic review." *Journal of Experimental Criminology* 10(4):399–428.

Hidalgo, Daniel and Benjamin Lessing. 2015. "Endogenous state weakness in violent democracies: paramilitaries at the polls." *Work. Pap., Mass. Inst. Technol., Cambridge, MA* .

Human Rights Watch. 2013. ""No Money, No justice": Police Corruption and Abuse in Liberia.".

International Crisis Group. 2009. *Liberia: Uneven Progress in Security Sector Reform*. Number 148 International Crisis Group.

Johnson, Devon. 2009. "Anger about crime and support for punitive criminal justice policies." *Punishment & Society* 11(1):51–66.

Jung, Danielle F and Dara Kay Cohen. 2020. *Lynching and local justice: Legitimacy and accountability in weak states*. Cambridge University Press.

Kagoro, Jude. 2019. "The Crime Preventers Scheme: A Community Policing Initiative for Regime Security in Uganda." *Journal of Intervention and Statebuilding* 13(1):41–56.

Karim, Sabrina. 2020. "Relational state building in areas of limited statehood: Experimental evidence on the attitudes of the police." *American Political Science Review* 114(2):536–551.

Krasner, Stephen D and Thomas Risse. 2014. "External actors, state-building, and service provision in areas of limited statehood: Introduction." *Governance* 27(4):545–567.

Mack, Andrew, Sebastian Merz, Mai Bui and Tara Cooper. 2013. *Human security report 2013: the decline in global violence: evidence, explanation, and contestation*. Human Security Report Project, Simon Fraser University.

Migdal, Joel S and Klaus Schlichte. 2005. "Rethinking the state." *The dynamics of states: The formation and crises of state domination* pp. 1–40.

Migdal, Joel Samuel, Atul Kohli and Vivienne Shue. 1994. *State power and social forces: domination and transformation in the Third World*. Cambridge University Press.

Moore, Mark Harrison. 1992. "Problem-solving and community policing." *Crime and justice* 15:99–158.

OECD. 2008. *OECD Handbook on Security System Reform: Supporting Security and Justice*. Organisation for Economic Co-operation and Development.

Ostrom, Elinor. 1996. "Crossing the great divide: coproduction, synergy, and development." *World development* 24(6):1073–1087.

Post, Alison E, Vivian Bronsoler and Lana Salman. 2017. "Hybrid regimes for local public goods provision: a framework for analysis." *Perspectives on Politics* 15(4):952–966.

Reeve, Richard and Jackson Speare. 2012. "Human security in Liberia. Local perspectives on formal and informal security sectors." *Accord: an international review of peace initiatives* 23.

Rose-Ackerman, Susan. 2004. *When states fail: causes and consequences*. Princeton University Press chapter Establishing the rule of law, pp. 182–221.

Rotberg, Robert. 2004. *The Failure and Collapse of Nation-States: Breakdown, Prevention, and Collapse*. Princeton, NJ: Princeton University Press chapter 1, pp. 201–213.

Sahin, Nusret, Anthony A. Braga, Robert Apel and Rod K. Brunson. 2017. "The Impact of Procedurally-Just Policing on Citizen Perceptions of Police During Traffic Stops: The Adana Randomized Controlled Trial." *Journal of Quantitative Criminology* 33(4):701–726.

Steinberg, Jonathan. 2008. *Thin Blue Line — the unwritten rules of policing South Africa*.

Tapscott, Rebecca. 2017. "Local security and the (un) making of public authority in Gulu, Northern Uganda." *African Affairs* 116(462):39–59.

Tapscott, Rebecca. 2021. "Vigilantes and the State: Understanding Violence through a Security Assemblages Approach." *Perspectives on Politics* pp. 1–16.

Tsai, Lily L. 2011. "Friends or foes? Nonstate public goods providers and local state authorities in nondemocratic and transitional systems." *Studies in Comparative International Development* 46(1):46–69.

Tyler, Tom R. and Jeffrey Fagan. 2008. "Legitimacy and Cooperation: Why Do People Help the Police Fight Crime in Their Communities?" *Ohio State Journal of Criminal Law* 6:231–275.

United Nations. 2016. *Guidelines for Police Operations in United Nations Peacekeeping Operations and Special Political Missions*. United Nations Department of Peacekeeping Operations.

United Nations. 2018. *Community-Oriented Policing in United Nations Police Operations*. United Nations Department of Peacekeeping Operations.

Wisler, Dominique and Ihekwoaba D Onwudiwe. 2008. "Community policing in comparison." *Police Quarterly* 11(4):427–446.

Zanker, Franzisca. 2017. "Moving beyond Hybridity: The Multi-Scalar Adaptation of Community Policing in Liberia." *Journal of Intervention and Statebuilding* 11(2):166–185.

Strengthening the Rule of Law through Community Policing: Evidence from Liberia

Online Appendix

A.1 INTERVENTION PAMPHLET

Figure A.1: Example of a pamphlet distributed during patrols and meetings

Police accountability and the Professional Standards Division (PSD) of the LNP

- The Professional Standards Division is a special unit of the LNP responsible for handling incidents of Police Misconduct.
- If you witness an instance of police misconduct, you should report it to the PSD. You may do so by calling 0770800XXX, or by visiting the PSD office in person at LNP headquarters.
- You can also report police misconduct to Victor X, Chief of Patrol for Greater Monrovia (0770800XXX), the Zone Commander for your Zone, or any officer of the LNP.
- If you still don't get a response, bring your complaint to a Call In Show on the radio.

Remember, *no police officer* has the right to:

- Solicit a bribe from you
- Make you pay a fine without issuing a you citation
- Intervene in civil cases like land disputes or money business – that one is for the courts.
- Use excessive force against civilians

Community Watch Teams

A Community Watch Team is a community group comprised of community leaders and volunteers chosen and vetted by the community and the police.

Responsibilities of the community watch team

1. Conduct night patrols in the community to deter criminal activity
2. Report all suspects immediately to the police
3. Provide information on suspected criminal activity to the LNP
4. Respect freedoms and rights of individuals arrested as violators or suspects
5. Protect the lives, security and properties of citizens and other persons in their respective communities

Rules governing the community watch team

1. Members MUST report all crimes to the Police
2. Members have arresting power but should report violators immediately to the Police
3. Membership is a Voluntary Service and not money making
4. Members should *never use weapons or engage in mob violence*
5. Members should not associate with known criminals or engage in collusion with said criminals

If your community would like to start a watch team, ask your Chairman/Chairlady contact the Community Services Section at the Zone Base.

Liberia National Police

Community Policing Program

POLICE ZONE 2

What is community policing?

Community policing is an approach to policing which recognizes the interdependence and shared the responsibility of the police and the community in ensuring a safe and secure environment.

Goals of community policing

- Prevent crime in the community through increased police presence and visibility
- Encourage communities to share information with the police
- Empower the community on issues of crime and their legal rights
- Promote open and honest relations between police and the residents
- Promote greater accountability in policing

Benefits associated with community policing

- Improved relationships between citizens and police
- Increased trust in police as well as information flow from the community
- Residents feel more comfortable approaching police officers due to greater familiarity
- Officers feel less hostility from the residents due to improved relations
- Greater police effectiveness due to improved team work between police and residents
- Greater police accountability

Women and Children Protection Services (WACPS)

- ✓ The Women and Children Protection Services (WACPS) division is trained to deal with cases of sexual and gender based violence, domestic abuse, and child abuse.
- ✓ If you know someone who is a victim of rape, domestic violence, child abuse, human trafficking, child marriage, or child labor, report it to WACPS
- ✓ Officers from WACPS are available at every police station. They have safe houses where victims can access food and shelter, and they can assist victims in getting help from other government agencies and NGOs, such as medical treatment, counseling, and legal assistance.

WACPS would like you to know:

- ✓ Sexual intercourse between someone over 18 and someone under 18 is called statutory RAPE. Statutory rape is a crime, just like forcible rape, and must be reported to the police.
- ✓ Be suspicious when you see uncles, step-fathers, neighbors, teachers, and mothers' boyfriends who are unusually friendly with young girls, as these people are the most common perpetrators of statutory rape.
- ✓ Child labor is also a common problem in Liberia. If you know of a child who has to work or sell instead of going to school, report it to WACPS

Key contacts at Zone 2 Police Base

Zone 2 Commander
Chief Superintendent James H. X
0770-800- XXX

Executive Officers / Patrol
Superintendent Veronica X
0770-800- XXX
Chief Inspector Sam B. X
0770-800- XXX

Chief of Operations
Chief Inspector Kamata T. X
0770-800- XXX

Community Services Section
Chief Inspector William R. X
0778-481- XXX / 0776-616- XXX
Inspector Edward X
0778-481- XXX / 0776-905- XXX
Sergeant Elaine T. Farley
0778-481- XXX

Women & Children Protection Services
Chief Inspector Anthony X
0770-800- XXX

Crime Services Department (CSD)
Inspector Edward X
0770-800- XXX
Sergeant J. Cyrus X
0770-800- XXX

Traffic Section
Inspector Muhammed X
0776-551-XXX

Radio Number: 0770-800-911

Other Important Numbers
____

Observation Notes from Town Hall Meetings

Messages communicated by police during town hall meetings

In addition to lectures on the importance of police - community partnerships, crime reporting, and the various services and divisions of the LNP, most meetings featured a lecture on the Watch Forum initiative. In the beginning, the focus was on introducing residents to the initiative and explaining how communities should go about organizing a Watch Forum.

As a research assistant wrote in his Observation Notes:

> [The Zone 9 Commander] lectured on organizing a Watch Forum. He discussed the definition of a Watch Forum and the roles and responsibilities of the Forum. The Watch Team should be composed of minimum 15 members based on the size of the community and the willingness of the citizens.
>
> . . . After the recruitment process, candidates will be vetted by Crime Services Division to ensure they are free of criminal charges. The leader of the Forum shall complete the Application form obtained from CPOs with attached passport size photos. Once approved they will be directed as to where they can obtain ID cards.
>
> . . . He educated the residents that the Watch Forum should report all suspected criminal activities to the police. Members may apprehend suspected criminals where necessary but should report said criminals immediately to the police. They should desist from Mob Action against any violator or suspected criminals. The should not apply mob justice. They are only the eyes and ears of the police. He concluded that the community and the police need to build an active partnership in the fight against crime.[10]

Despite the program's emphasis on the formal application and vetting process, these procedures were seldom followed. Communities struggled to raise funds for passport photos and ID cards, while the LNP was never able to produce the official application form and almost certainly would not have been able to check candidates' criminal records.[11] In practice, vetting was the responsibility of the Town Chairman/woman and whomever s/he assigned to lead the Watch Forum. In many communities, members were drawn from pre-existing security groups that had been operating independently of the police. Training was minimal, and consisted mainly of lectures de-

[10]Ballah Creek Community, 4/14/2018.

[11]To the best of my knowledge, the LNP does not maintain a database that would allow for this.

livered either as part of the intervention or through separate security meetings organized between CPOs and Forum leaders.

But if there was one thing that officers *were* consistent about, it was in warning residents to refrain from mob violence or any form of force. Members of the watch team "should be persuasive and use good manner of approach when interacting with people," an officer told attendees in Neckly Town, "they may not and should not carry weapons because they are not vigilantes."[12] In another community, the Commander made clear that instances of force by Watch Forums were against the law and would be prosecuted: "Remember, in as much as we are encouraging you to form the Watch Forum, you will answer to the law for any act of violence you commit."[13] The message was much the same in Lajoy Community: "If you bring a rogue to the police station and he is hurt and bloody, we will detain you and hold you responsible."[14] In addition to the threat of sanction, the officers also appealed to principles of human rights, the protections enshrined in Liberian law, and the importance of due process in trying to dissuade residents from mob violence.

The threat of sanction was not the only argument that officers made to dissuade residents from mob violence; they also appealed to principles of human rights, the protections enshrined in Liberian law, and the importance of due process. The following exchanges are illustrative:

> Question: Participant Solomon asked why criminal should not be beaten when he is rightfully caught committing a crime?
>
> Answer: Patrolman Henry Flomo responded that beating the criminal is *prejudging* and torturing him. The law of Liberia is against torture. The law also says that all suspects are innocent until proven guilty in a court of law. Therefore, the suspect must be turned over to the police for investigation and eventual prosecution.

And:

> Question: Participant Ramcy asked, "suppose a criminal enters my house, and in the process of tussling he gets killed, will I be right?"
>
> Answer: Patrolman Henry Flomo responded that Murder is a crime and it is punishable by law. Report all suspected criminals to the police for investigation and allow

[12] Neckly Town, 5/6/2018.
[13] Jamaica Road, 6/16/2018.
[14] Lajoy Community, 9/9/2018.

the law to take its course.

Later, in the course of the intervention the focused was on encouraging residents to be proactive in organizing and sustaining Watch Forums:

> [The commander] said citizens have a social responsibility to establish Watch Forums in every community to work in collaboration with the Liberian National Police as a way of crime prevention at the community level. He also encouraged the citizens to actively participate in community policing activities and decision making that affects the security of their community. ... if a crime is being committed where police officers are absent, the Watch Forum will help to arrest the suspect and immediately turn him/ her over to the nearest police depot/station without harm on the suspect or call the number on the pamphlet. The patrolman said "Remember, you will answer to the law for any act of violence you commit in as much as we encouraging you to form the Watch Forum."[15]

RESIDENTS DESCRIBING ACTIONS TAKEN TO PREVENT VIGILANTISM

Observation Notes recorded by local research assistants during the town hall meetings document several instances in which residents spoke of local security groups diffusing situations that otherwise could have escalated into mob violence. In one community, the Chairman recounted how members of the newly-initiated Watch Forum performed a citizens' arrest of a suspicious person in the middle of the night and reported him to the nearest police station — a situation which before the intervention, may have led to rough-handling of the suspect or worse.[16] In another, a member of the Watch Forum recounted how he and his colleagues directly intervened to prevent mob violence:

> The Watch Forum member said that he was grateful to the police for the knowledge they have gained from the Watch Forum initiative. He said that recently the Watch Forum was able to save a stranger in their community from mob violence. He said there was a strange man who could not speak to the common understanding of the local residents. Community people had surrounded him to beat him because they suspected him to be a missing criminal.
>
> According to him, because of their knowledge on mob violence and their role in the Watch Forum, they were able to intervene and asked the man what he wanted to do in their community? The man told them that he went to see somebody in the community

[15] Jamaica Road, 06/16/2018.
[16] Kpelleh Town Back Road, 8/11/2018.

> but he forgot the person's name. They also asked him about where he came from? He told them he came from a church and they decided to walk with him to the church. When they got to the church, they ask the people at the church whether they knew him? The people at church said yes they knew the man and they thanked them for protecting the man.[17]

References to members of the Watch Forum escorting suspects to the police or otherwise diffusing dangerous situations came up in at least two other town hall meetings.[18] Though rare as a proportion of the total number of meetings, neither the meeting agenda nor the observation protocol were designed to screen for these events, and it is likely that others went unrecounted or unrecorded.

[17] Catholic Community, 9/23/2018.

[18] Nyanford Community, 10/21/2018; New Hope Community, 8/4/2018.

A.2 Research ethics and minimizing risks to human subjects

The research team took extraordinary efforts ensure that this research project adhered to the Belmont principles for ethical research. IRB approval for this study was obtained from [REDACTED FOR PEER REVIEW]. Research approval was also obtained from the Inspector General of the Liberian National Police and the town chairperson of every community in the study sample. Voluntary and informed consent was obtained verbally from all participants in the baseline and endline survey. All participants were informed about the research purpose of the study, and no deception was used.

Going beyond these basic requirements, the research team took several steps to minimize risks. First, prior to the intervention, we carefully vetted the police officers with whom we worked. We did this by getting to know the Community Policing Officers (CPOs) over several meetings prior to the start of the intervention, and by informally checking their backgrounds and credentials with other trusted police officers with whom we already had a relationship. (At the outset of this study, the research team had been working closely with the LNP for three years through collaboration on prior research studies).

The second step we took was to embed a local research assistant within the police as an independent observer during every town hall meeting and foot patrol conducted for this study. This research assistant also established an independent line of communication with a diverse set of community leaders in every treatment community, including the town chairperson, the women's group leader, the youth group leader, and the minority group leader (where present). Throughout the intervention, research assistants touched base with community leaders regularly both in person and by phone to inquire about any potentially adverse events or risks.

The final step we took to minimize risks was to pre-determine a number of actions that we were prepared to take in response to an adverse event. Depending on the nature of the event, if it involved police misconduct, we were well-positioned to report such misconduct to Zone Commanders, or to higher authorities at the LNP's Central Headquarters. Another option we identified was to drop a particular Police Zone from the study altogether, if it appeared that misconduct was isolated to officers in that particular zone, or if the Zone Commander was not prepared to take corrective action (because we block randomized within police zones, we would have been able to drop entire police zones without the risk of bias). And if our back-channel communications suggested that communities as a whole were responding adversely to the intervention, we were prepared to stop the intervention altogether in that particular community, and to ramp up monitoring in other communities.

Our monitoring efforts did not reveal any adverse events during this study.

In terms of the maximizing research benefits, the research team viewed this study as an opportunity to advance a locally-grown community policing program with the potential to deliver substantial benefits to citizens. To this end, results from this study were presented to the LNP's leadership as well as to CPOs at each participating station. The research team also presented to national and international NGOs in Liberia working on security sector reform. And lastly, the research team produced a policy memo for distribution to these practitioners and has sought publication in influential journals read by security sector reform policymakers.

A.3 SAMPLING AND RANDOMIZATION

A complete description of the sampling and randomization procedure for this study is as follows: Within each of Monrovia's ten police zones, local research assistants worked with the CPOs to identify any 'high priority' communities to be nominated for the intervention based on their assessments of crime rates, police-community relations, or other factors. This process identified 35 'high priority' communities. Because this sample size would have resulted in an under-powered study, an additional 65 communities were randomly sampled from the remaining population of communities to yield a target sample size of 100 communities.[19] The number of additional communities sampled per zone was proportional to the total number of communities in each zone.

Half of the sampled communities within each zone were then randomly assigned to treatment via block randomization. In zones with an odd number of communities, I randomly assigned $(N_b - 1)/2$ communities to treatment, where N_b denotes the number of communities in block b, resulting in a slightly less or slightly higher than .5 probability of assignment to treatment, depending on rounding. I account for this in the analysis by weighting observations by the inverse of the probability of assignment to treatment/control, following Gerber and Green (2012, 117), and as described in the Estimation section of the paper.

During the baseline survey, two communities were found to be duplicates of other communities and were dropped; during implementation, staffing constraints within the implementation monitoring team required that the smallest police zone be dropped. These two changes resulted in a final sample size of 93 communities, 45 of which were assigned to treatment.

SAMPLING RESPONDENTS WITHIN COMMUNITIES

Communities in Monrovia are sub-divided into anywhere from four to 7 'blocks', which are akin to small neighborhoods or street blocks in the United States. Because many communities as a

[19] Based on a sample size of 100 communities, 20 observations per community, and an intra-cluster correlation of .1, we estimated that we the study would detect effects as small as .18 standard deviations with 80% power.

whole are very large, with populations of up to 5000 residents, the intervention targeted the most central block in each community plus the largest two adjacent blocks. Within each community, respondents for the baseline and endline surveys were randomly sampled from the selected blocks following a random walk procedure. The same sampling procedure focused on the most central block and the largest two adjacent blocks was followed in both treatment and control communities. The baseline and endline surveys were repeated cross-sections of the communities rather than a panel survey of the same individuals at baseline and endline. For this reason, concerns about attrition are not applicable.

This sampling strategy is accounted for in the analysis by weighting observations by the inverse of the probability that an individual was selected for the endline survey. In calculating these weights, I make the simplifying assumption all blocks within a community are of the same size, and calculate the sampling probability for individual i in community c as: $\frac{1}{3 \times \frac{Town\_Pop_c}{Num\_Blocks_c}}$.

RISK OF SPILLOVER BETWEEN SAMPLED COMMUNITIES

There are two potential sources of spillover in this study. The first is the risk that treatment spills-over into nearby control communities, partially treating control residents. This could occur either because residents of control communities hear about and attend meetings in treatment communities, or because residents of treatment communities share what they learn with residents of nearby control communities. Although plausible, several considerations suggest the risk of this type of spillover is low. Communities in my sample are large and populous, with an average population of 6,126. No two communities in the sample are adjacent, and the average distance of control respondents to the nearest treated respondent at endline is .36 kilometers, a large and meaningful distance in the context of Monrovia's densely populated communities. There is also little evidence that residents of control communities attended meetings in treatment communities: at baseline, 7% of residents in control communities attended a meeting with police in the past six months; at endline, this number rose to only 10%. Meanwhile, meeting attendance in treatment communities went from 7% at baseline to 42% at endline.

The second source of spillover derives from the randomization procedure, which randomizes within but not across police zones/jurisdictions. This means that officers who participate in the intervention will also be interacting with residents of control communities in the course of their regular, everyday responsibilities. If the intervention leads to changes in officers' that in turn influence residents' perceptions and behaviors in control communities, then this could lead me to underestimate the program's impact. For instance, the program could in theory make officers more empathetic or caring, which in turn could improve residents perceptions in control communities.

My study design cannot account for these kinds of 'general equilibrium' dynamics. For this reason, my results should be interpreted as the 'partial equilibrium' effects of changing citizens' attitudes of and exposure to the police while holding constant any resulting changes in police behavior, which would have influenced treatment and control communities equally.

A.4 ENDLINE DESCRIPTIVE STATISTICS

Table A.1: Descriptive statistics

	Mean	N
Demographics		
Male	0.48	1,851
Age	37.3	1,851
Christian	0.88	1,851
No education	0.12	1,851
Elementary education	0.063	1,851
Junior High education	0.19	1,851
High schools education	0.42	1,851
Post secondary education	0.21	1,851
Literate	0.60	1,851
Familarity w/ police		
Knows about WACPS	0.94	1,851
Knows about PSD	0.87	1,851
Knows about CSD	0.40	1,851
Knows WACPS by name	0.55	1,851
Knows PSD by name	0.25	1,851
Knows local commander	0.20	1,851
Knows officer by name	0.40	1,851
Knows officer's number	0.21	1,851
Knowledge of law		
Knows statutory rape illegal	0.91	1,851
Knows about habeas corpus	0.94	1,851
Knows right to lawyer	0.91	1,851
Knows rights when reporting	0.96	1,851
Knows about child support	0.92	1,851
Knows obligation to report serious crimes	0.34	1,851
Knows informal fees illegal	0.32	1,851
Knows bond fees illegal	0.31	1,851
Police intentions		
Police corrupt?	0.48	1,851
Police treat all equal	0.42	1,851
Police take cases seriously	0.67	1,851
Police fair to all sides	0.54	1,851
Police care about citizens' safety	0.69	1,851
Police respect citizens	0.56	1,851
Police objective	0.62	1,851
Police respect victims	0.70	1,851

Notes: Apart from *Age*, all variables are coded dichotomously.

Table A.2: Descriptive statistics (cont.)

	Mean	N
Cooperation norms		
Ppl get mad if you report a burglary?	0.28	1,851
Ppl get mad if you report domestic violence?	0.43	1,851
Ppl get mad if you report land disputes?	0.29	1,851
Ppl should always obey the police	0.33	1,851
Ppl get mad if you give info about armed robbery?	0.26	1,851
Ppl get mad if you give info about domestic violence?	0.30	1,851
Ppl get mad if you give info about stolen motorcycle?	0.23	1,851
Ppl get mad if you give info about child abuse?	0.30	1,851
Trust in police capacity		
Police able to respond quickly	0.62	1,851
Police able to investigate effectively	0.80	1,851
Police act on citizens' feedback	0.60	1,851
Police include citizens in decision making	0.75	1,851
Support for mob violence		
Mob violence justified for rape?	0.16	1,851
Mob violence justified for armed robbery?	0.22	1,851
Mob violence justified for burglary?	0.19	1,851
Community coproduction of security		
Seen security mtg past month	0.43	1,849
Attended security mtg past month	0.24	1,850
Seen patrol past month	0.39	1,848
Attended patrol past month	0.19	1,851
Gave food/tea past month	0.32	1,850
Town has Watch Forum?	0.27	1,850
Forum patrols at night?	0.22	1,851
Forum meetings regularly?	0.18	1,851
Forum registered with police?	0.23	1,848
Knowledge of watch team rules		
Weapons prohibited	0.23	1,851
Physical harm prohibited	0.73	1,851
Must avoid risks & danger	0.37	1,851
Checkpoints prohibited	0.17	1,851
Only operate in home community	0.49	1,851
Can perform citizens' arrest	0.95	1,851

Notes: All variables are coded dichotomously.

Table A.3: Descriptive statistics (cont.)

	Mean	N
Prefer police respond to:		
Land disputes	0.27	1,851
Violent land disputes	0.63	1,851
Burglaries	0.80	1,851
Domestic violence	0.44	1,851
Armed robbery	0.88	1,851
Crime tips & information sharing		
Reported suspicious activity past 6m	0.091	1,851
Helped police find suspect past 6m	0.068	1,849
Given info for investigation past 6m	0.097	1,849
Provided testimony past 6m	0.068	1,851
Willingness to report police misconduct		
Would report illegal checkpoint	0.68	1,849
Would report drunk officer	0.49	1,849
Would report police beating	0.85	1,851
Crime		
# of armed robberies	0.41	1,844
# burglaries	1.39	1,833
# of aggravated assaults	0.48	1,840
# simple assaults	0.65	1,837
# sexual violence	0.069	1,850
# domestic violence	0.64	1,842
# domestic violence	0.91	1,817
# violence land disputes	0.26	1,846
# non violent land disputes	0.20	1,817
# other violent crimes	0.027	1,803
# other non violent	0.069	1,818
# murders	0.050	1,851
# child abuse	0.32	1,838
Perceptions of security		
Fears violent crime	0.52	1,851
Fears non-violent crime	0.53	1,851
Fears walking at night	0.34	1,851
Fears home invasions	0.27	1,851
House boundaries secure	0.58	1,851
House items secure	0.45	1,851
Motorbike secure outside	0.093	1,851
Generator secure outside	0.076	1,851
Overall satisfaction with police performance		
Trust police?	0.62	1,851
Satisfied w/ police performance?	0.61	1,851
Mob violence		
# of incidents in community (past 12 months)	0.75	1,851

Notes: Apart from crime variables, all variables are coded dichotomously.

A.5 TESTS OF BALANCE

Table A.4 shows the estimated mean for the control group (Column 2) and the estimated difference in means between the treatment and control groups. The estimates are produced using WLS regression of each outcome on treatment and block fixed effects to account for block randomization, with weights accounting for unequal probabilities of assignment.

Table A.4: Tests of balance

	(1) T-C Difference in means	(2) Control mean
male	0.01	0.44***
	(0.03)	(0.02)
age	-1.14	37.42***
	(0.78)	(0.56)
rel_Christian	0.03	0.87***
	(0.03)	(0.02)
education	0.07	4.36***
	(0.16)	(0.11)
literate	0.01	0.62***
	(0.03)	(0.02)
know_pol_idx	-0.05	0.02
	(0.08)	(0.06)
know_idx_lbr	0.01	-0.00
	(0.05)	(0.04)
intentions_idx_lbr	0.06	-0.03
	(0.07)	(0.04)
norm_idx_lbr	-0.06	0.03
	(0.06)	(0.03)
police_capacity_idx	0.04	-0.02
	(0.07)	(0.04)
responsive_act	0.01	-0.00
	(0.05)	(0.04)
ca_sec_idx	-0.01	0.00
	(0.10)	(0.07)
sup_mobviol_idx	0.12*	-0.06
	(0.06)	(0.04)
crimeres_idx_lbr	-0.08	0.04
	(0.05)	(0.05)
crime_tips_idx_lbr	0.02	-0.01
	(0.06)	(0.04)
police_abuse_report_idx_lbr	0.03	-0.02
	(0.07)	(0.06)
crime_num_lbr	0.16*	-0.08**
	(0.07)	(0.03)
cmob_num	0.03	-0.02
	(0.08)	(0.04)
future_security_idx_lbr	-0.02	0.01
	(0.07)	(0.04)
N		

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

A.6 Intensity of treatment: full regression results

Table A.5 reports the full results from Table 2 in the main text, "Intensity of treatment".

Table A.5: Average treatment effects on intensity of treatment outcomes

	(1) Intensity of treatment index	(2) Sees foot patrols daily or weekly	(3) Sees vehicle patrols daily or weekly	(4) Attended police mtg past 6m
treatment	0.42***	0.01	0.02	0.31***
	(0.06)	(0.02)	(0.02)	(0.03)
cb_compliance_idx	0.09	0.01	-0.01	0.06
	(0.11)	(0.04)	(0.03)	(0.07)
Constant	-0.21*	0.08*	0.08*	-0.06
	(0.09)	(0.03)	(0.04)	(0.04)
Ctrl_mean	0.06	0.11	0.12	0.10
Observations	1836	1851	1851	1851

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$

A.7 FIGURE 1 IN TABLE FORMAT: EFFECTS ON HYPOTHESIZED MECHANISM INDICES

Tables A.6 and A.7 report the results from Figure 1 in the main text, “Effects on hypothesized mechanisms” in regression table format.

Table A.6: Average treatment effects on hypothesized mechanisms

	(1) Familiarity w. police	(2) Knowledge of law	(3) Perceptions of police intentions	(4) Norms supporting cooperation	(5) Perceptions of police capacity
treatment	-0.03	-0.02	0.09*	0.04	0.11*
	(0.03)	(0.03)	(0.04)	(0.04)	(0.05)
cb_know_pol_idx	-0.01				
	(0.07)				
cb_know_idx_lbr		0.04			
		(0.11)			
cb_intentions_idx_lbr			-0.09		
			(0.11)		
cb_norm_idx_lbr				-0.03	
				(0.12)	
cb_police_capacity_idx					-0.12
					(0.10)
_cons	-0.45***	-0.11*	0.34***	0.09	0.48***
	(0.06)	(0.05)	(0.08)	(0.05)	(0.10)
N	1851	1851	1631	1741	1806
Block fixed effects	Y	Y	Y	Y	Y

Each outcome is standardized by baseline mean. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

Table A.7: Average treatment effects on hypothesized mechanisms (continued)

	(1) Perceptions of police responsiveness	(2) Contributions to community coproduction	(3) Support for mob violence	(4) Knowledge of watch forum rules
treatment	0.03	0.07*	-0.07+	0.07**
	(0.06)	(0.03)	(0.04)	(0.02)
cb_pol_responsiveness_lbr	-0.02			
	(0.10)			
cb_ca_sec_idx		0.09+		
		(0.05)		
cb_sup_mobviol_idx			0.15	
			(0.11)	
cb_know_cwt_idx				-0.07
				(0.08)
_cons	0.07	0.16***	-0.43*	-0.27***
	(0.11)	(0.04)	(0.18)	(0.05)
N	1786	1841	1831	1851
Block fixed effects	Y	Y	Y	Y

Each outcome is standardized by baseline mean. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

A.8 Figure 2 in table format: effects on coproduction disaggregated by component variables

Tables A.8 to A.11 report the results from Figure 2 in the main text, "Effects coproduction disaggregated by component variables" in regression table format.

Table A.8: Average treatment effects on hypothesized mechanisms

	(1) Seen sec mtg past month	(2) Attended mtg past month	(3) Attended patrol past month	(4) Seen patrol past month	(5) Gave food or tea past month
treatment	0.16***	0.12***	0.06	0.05+	0.02
	(0.04)	(0.03)	(0.05)	(0.03)	(0.04)
cb_ca_sec_idx	0.06	0.02	0.14*	0.05	0.09
	(0.06)	(0.05)	(0.06)	(0.04)	(0.06)
_cons	0.25***	-0.01	0.25***	0.12**	0.23***
	(0.07)	(0.05)	(0.06)	(0.04)	(0.06)
N	1849	1850	1848	1851	1850
Block fixed effects	Y	Y	Y	Y	Y

Each outcome is standardized by baseline mean. Cluster robust standard errors in parenthesis, clustered by community. + $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

Table A.9: Average treatment effects on hypothesized mechanisms

	(1) Town has Watch Forum?	(2) Forum patrols at night?	(3) Forum meets regularly?	(4) Forum registered with police?	(5) Mob viol justified for rape?
treatment	0.06	0.05	0.05	0.07	-0.02
	(0.05)	(0.05)	(0.04)	(0.05)	(0.02)
cb_ca_sec_idx	0.13^{+}	0.13^{*}	0.09^{+}	0.13^{+}	
	(0.07)	(0.06)	(0.05)	(0.07)	
cb_sup_mobviol_idx					0.08
					(0.05)
_cons	0.22^{***}	0.16^{**}	0.13^{**}	0.12^{*}	0.12
	(0.06)	(0.05)	(0.05)	(0.06)	(0.08)
N	1850	1851	1851	1848	1851
Block fixed effects	Y	Y	Y	Y	Y

Each outcome is standardized by baseline mean. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

Table A.10: Average treatment effects on hypothesized mechanisms

	(1) Mob viol justified for armed robbery?	(2) Mob viol justified for burglary?	(3) Weapons prohibited	(4) Physical harm prohibited	(5) Members must avoid risks & danger
treatment	-0.04**	-0.03	0.08***	0.05*	0.05^{+}
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
cb_sup_mobviol_idx	0.07	0.07			
	(0.05)	(0.06)			
cb_know_cwt_idx			-0.03	0.00	-0.05
			(0.04)	(0.06)	(0.07)
_cons	0.23**	0.17^{+}	-0.03	0.46***	0.27***
	(0.08)	(0.09)	(0.05)	(0.05)	(0.05)
N	1851	1851	1851	1851	1851
Block fixed effects	Y	Y	Y	Y	Y

Each outcome is standardized by baseline mean. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

Table A.11: Average treatment effects on hypothesized mechanisms

	(1) Checkpoints prohibited	(2) Members can only operate in home community	(3) Members can perform citizens' arrest
treatment	0.02	-0.01	0.02*
	(0.02)	(0.04)	(0.01)
cb_know_cwt_idx	-0.06	-0.06	0.01
	(0.05)	(0.11)	(0.03)
_cons	-0.01	0.31***	0.90***
	(0.05)	(0.06)	(0.03)
N	1851	1851	1851
Block fixed effects	Y	Y	Y

Each outcome is standardized by baseline mean. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

A.9 FIGURE 3 IN TABLE FORMAT: EFFECTS ON PRIMARY OUTCOMES WITH MULTIPLE COMPARISONS ADJUSTMENTS

Tables A.12 and A.13 reports the results on this study's primary outcome indices with p-values adjusted for multiple comparisons following Benjamini and Hochberg (1995). After adjusting the p-values, the negative effect on incidence of mob violence remains significant at the 1% level; likewise, effects on all of the other outcomes remain not statistically significant, as reported in the paper.

Table A.12: Average treatment effects on primary outcomes with multiple comparisons adjustments

	(1) Willingness to report crimes	(2) Crime tips & info sharing	(3) Willingness to report misconduct	(4) Total # of crimes reported
treatment	0.01	-0.03	0.00	0.01
	(0.04)	(0.04)	(0.04)	(0.16)
cb_crimeres_idx	0.09			
	(0.09)			
cb_crime_tips_idx		0.03		
		(0.08)		
cb_police_abuse_report_idx			0.03	
			(0.08)	
cb_crime_num				0.75
				(0.61)
_cons	0.13^{+}	-0.26^{***}	-0.04	0.39
	(0.07)	(0.06)	(0.08)	(0.38)
N	1851	1847	1823	1778
P value	0.70	0.47	0.92	0.94
BH P value	0.93	0.85	0.93	0.93
Block fixed effects	Y	Y	Y	Y

Each outcome is standardized by baseline mean. Cluster robust standard errors in parenthesis, clustered by community. P-values adjusted following Benjamini and Hochberg (1995). $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

Table A.13: Average treatment effects on primary outcomes with multiple comparisons adjustments (continued)

	(1) Perceptions of security	(2) Satisfaction w/ police performance	(3) Incidence of mob violence
treatment	0.03	0.05	-0.28**
	(0.05)	(0.05)	(0.08)
cb_future_insecurity_idx	-0.12		
	(0.11)		
cb_satis_idx		-0.08	
		(0.11)	
cb_cmob_num			0.21*
			(0.10)
_cons	0.09	0.60***	0.59**
	(0.08)	(0.11)	(0.22)
N	1848	1839	1841
P value	0.50	0.28	0.00
BH P value	0.85	0.85	0.01
Block fixed effects	Y	Y	Y

m

Each outcome is standardized by baseline mean. Cluster robust standard errors in parenthesis, clustered by community. P-values adjusted following Benjamini and Hochberg (1995). $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

A.10 EFFECTS ON INDICES AND COMPONENT DEPENDENT VARIABLES

The following sub-sections estimate treatment effects for each outcome index and its component variables, following the estimation procedure described in Section 5.3 of the main paper.

FAMILIARITY WITH POLICE

Table A.14: Average treatment effects on familiarity with police

	(1) Familiarity with police index	(2) Knows about WACPS	(3) Knows about PSD	(4) Knows about CSD	(5) Knows WACPS by name
treatment	-0.03	-0.02*	-0.01	-0.05	-0.08*
	(0.03)	(0.01)	(0.02)	(0.04)	(0.03)
cb_know_pol_idx	-0.01	-0.02	0.00	-0.07	-0.14+
	(0.07)	(0.03)	(0.04)	(0.07)	(0.07)
_cons	-0.45***	0.87***	0.83***	0.09	0.13*
	(0.06)	(0.03)	(0.03)	(0.06)	(0.06)
N	1851	1851	1851	1851	1851
P value	0.38	0.04	0.67	0.25	0.02
BH P value	NA	0.08	0.67	0.29	0.08

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

Table A.15: Average treatment effects on familiarity with police (continued)

	(1) Knows PSD by name	(2) Knows Commander by name	(3) Knows officer by name	(4) Knows officer's number
treatment	-0.07*	0.04*	0.04	0.02
	(0.03)	(0.02)	(0.03)	(0.02)
cb_know_pol_idx	-0.06	0.08+	0.12*	0.02
	(0.07)	(0.04)	(0.05)	(0.05)
_cons	0.06	0.07+	0.28***	-0.01
	(0.05)	(0.04)	(0.05)	(0.04)
N	1851	1851	1851	1851
P value	0.03	0.04	0.12	0.22
BH P value	0.08	0.08	0.19	0.29

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

KNOWLEDGE OF LAW

Table A.16: Average treatment effects on knowledge of law

	(1) Knowledge of law index	(2) Knows statutory rape illegal	(3) Knows about habeas corpus	(4) Knows right to lawyer	(5) Knows rights when reporting
treatment	-0.02	0.00	-0.02^{+}	-0.01	-0.01
	(0.03)	(0.01)	(0.01)	(0.01)	(0.01)
cb_know_law_idx	0.04	-0.01	0.00	-0.00	-0.03
	(0.10)	(0.05)	(0.04)	(0.04)	(0.03)
_cons	-0.11*	0.83***	0.90***	0.82***	0.97***
	(0.05)	(0.03)	(0.03)	(0.03)	(0.03)
N	1851	1851	1851	1851	1851
P value	0.46	0.76	0.07	0.36	0.30
BH P value	NA	0.88	0.44	0.72	0.72

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

Table A.17: Average treatment effects on knowledge of law

	(1) Knows about child support	(2) Knows obligation to report	(3) Knows informal fees illegal	(4) Knows bond fees illegal
treatment	-0.03	-0.00	0.01	0.01
	(0.02)	(0.03)	(0.03)	(0.03)
cb_know_law_idx	0.05	0.17^{+}	0.01	-0.07
	(0.06)	(0.09)	(0.10)	(0.09)
_cons	0.88^{***}	0.38^{***}	0.05	0.12^{*}
	(0.03)	(0.05)	(0.05)	(0.05)
N	1851	1851	1851	1851
P value	0.11	0.88	0.80	0.83
BH P value	0.44	0.88	0.88	0.88

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

Perceptions of police intentions

Table A.18: Average treatment effects on perceptions of police intentions

	(1) Police intentions Index	(2) Police corrupt?	(3) Police treat all equal?	(4) Police take cases seriously?	(5) Police fair to all sides?
treatment	0.09*	-0.06*	0.06*	0.04+	0.01
	(0.04)	(0.02)	(0.03)	(0.02)	(0.02)
cb_intentions_idx	-0.08	-0.02	-0.02	-0.02	-0.07
	(0.12)	(0.05)	(0.06)	(0.07)	(0.06)
_cons	0.33***	0.55***	0.54***	0.73***	0.58***
	(0.08)	(0.05)	(0.05)	(0.06)	(0.06)
N	1631	1851	1851	1851	1851
P value	0.04	0.01	0.04	0.08	0.81
BH P value	NA	0.08	0.08	0.12	0.81

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

Table A.19: Average treatment effects on perceptions of police intentions (continued)

	(1) Police care about citizens' safety?	(2) Police respect citizens?	(3) Police objective?	(4) Police respect victims?
treatment	0.06*	0.04+	0.03	0.05*
	(0.03)	(0.02)	(0.02)	(0.02)
cb_intentions_idx	-0.05	-0.07	-0.07	-0.03
	(0.06)	(0.06)	(0.06)	(0.06)
_cons	0.71***	0.71***	0.73***	0.74***
	(0.06)	(0.05)	(0.05)	(0.05)
N	1851	1851	1851	1851
P value	0.03	0.09	0.20	0.02
BH P value	0.08	0.12	0.23	0.08

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

NORMS SUPPORTING COOPERATION

Table A.20: Average treatment effects on norms supporting cooperation

	(1) Cooperation norms index	(2) People get made if you report: A burglary?	(3) Domestic violence?	(4) Land disputes?	(5) Ppl should always obey the police
treatment	0.05	-0.02	-0.06*	-0.02	0.02
	(0.04)	(0.03)	(0.03)	(0.03)	(0.02)
cb_norm_idx	0.06	-0.04	-0.16*	-0.05	-0.05
	(0.12)	(0.09)	(0.07)	(0.08)	(0.07)
_cons	0.08	0.32***	0.52***	0.34***	0.53***
	(0.05)	(0.05)	(0.05)	(0.05)	(0.06)
N	1741	1851	1851	1851	1851
P value	0.19	0.54	0.02	0.39	0.29
BH P value	NA	0.57	0.16	0.64	0.66

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

Table A.21: Average treatment effects on norms supporting cooperation (continued)

	(1)	(2)	(3)	(4)
	People get made if you give info about:			
	Armed robbery?	Domestic violence?	Stolen motorcycle?	Child abuse?
treatment	-0.02	-0.04	-0.00	-0.02
	(0.03)	(0.03)	(0.03)	(0.03)
cb_norm_idx	-0.02	-0.06	-0.02	-0.05
	(0.11)	(0.09)	(0.09)	(0.09)
_cons	0.20***	0.25***	0.17***	0.24***
	(0.05)	(0.05)	(0.05)	(0.05)
N	1851	1851	1851	1851
P value	0.49	0.10	0.99	0.56
BH P value	0.64	0.40	0.99	0.64

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

PERCEPTIONS OF POLICE CAPACITY

Table A.22: Average treatment effects on perceptions of police capacity

	(1) Police capacity index	(2) Police able to respond quickly?	(3) Police able to investigate effectively?
treatment	0.11*	0.07**	0.02
	(0.05)	(0.03)	(0.02)
cb_police_capacity_idx	-0.12	-0.05	-0.05+
	(0.10)	(0.06)	(0.03)
_cons	0.48***	0.79***	0.93***
	(0.10)	(0.05)	(0.05)
N	1806	1851	1851
P value	0.03	0.01	0.33
BH P value	NA	0.02	0.33

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

PERCEPTIONS OF POLICE RESPONSIVENESS

Table A.23: Average treatment effects on perceptions of police responsiveness

	(1) Police responsiveness index	(2) Police act on citizens' feedback?	(3) Police include citizens in decision making?
treatment	0.03	-0.01	0.01
	(0.06)	(0.03)	(0.03)
cb_pol_responsiveness_lbr	-0.02	0.08*	0.01
	(0.10)	(0.04)	(0.05)
_cons	0.07	0.74***	0.73***
	(0.11)	(0.06)	(0.06)
N	1786	1851	1851
P value	0.68	0.83	0.64
BH P value	NA	0.83	0.83

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

CONTRIBUTIONS TO COMMUNITY COPRODUCTION

Table A.24: Average treatment effects on contributions to community coproduction

	(1) Coproduction contributions index	(2) Seen security meeting past month	(3) Attended security meeting past month	(4) Seen CWF patrol past month	(5) Attended CWF patrol past month
treatment	0.07*	0.16***	0.12***	0.06	0.05+
	(0.03)	(0.04)	(0.03)	(0.05)	(0.03)
cb_ca_sec_idx	0.09+	0.06	0.02	0.14*	0.05
	(0.05)	(0.06)	(0.05)	(0.06)	(0.04)
_cons	0.16***	0.25***	-0.01	0.25***	0.12**
	(0.04)	(0.07)	(0.05)	(0.06)	(0.04)
N	1841	1849	1850	1848	1851
Ctrl group mean	-0.10	0.35	0.18	0.36	0.17
P value	0.04	0.00	0.00	0.17	0.08
BH P value	NA	0.00	0.00	0.27	0.24

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. + $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

Table A.25: Average treatment effects on contributions to community coproduction (continued)

	(1) Gave food / tea past month	(2) Town has watch forum?	(3) Forum patrols at night?	(4) Forum meets regularly?	(5) Forum registered with police?
treatment	0.02	0.06	0.05	0.05	0.07
	(0.04)	(0.05)	(0.05)	(0.04)	(0.05)
cb_ca_sec_idx	0.09	0.13+	0.13*	0.09+	0.13+
	(0.06)	(0.07)	(0.06)	(0.05)	(0.07)
_cons	0.23***	0.22***	0.16**	0.13**	0.12*
	(0.06)	(0.06)	(0.05)	(0.05)	(0.06)
N	1850	1850	1851	1851	1848
Ctrl group mean	0.32	0.25	0.19	0.16	0.19
P value	0.62	0.21	0.25	0.19	0.11
BH P value	0.62	0.27	0.28	0.27	0.25

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. + $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

SUPPORT FOR MOB VIOLENCE

Table A.26: Average treatment effects on support for mob violence

	(1)	(2)	(3)	(4)
	Support for mob		Mob violence justified for:	
	violence index	rape?	armed robbery?	burglary?
treatment	-0.07^{+}	-0.02	-0.04**	-0.03
	(0.04)	(0.02)	(0.02)	(0.02)
cb_sup_mobviol_idx	0.15	0.08	0.07	0.07
	(0.11)	(0.05)	(0.05)	(0.06)
_cons	-0.43*	0.12	0.23**	0.17^{+}
	(0.18)	(0.08)	(0.08)	(0.09)
N	1831	1851	1851	1851
P value	0.12	0.33	0.01	0.17
BH value	NA	0.33	0.03	0.25

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

KNOWLEDGE OF WATCH FORUM RULES

Table A.27: Average treatment effects on knowledge of rules governing local security groups

	(1) Knowledge of security group rules index	(2) Weapons prohibited	(3) Physical harm prohibited	(4) Must avoid danger
treatment	0.07**	0.08***	0.05*	0.05+
	(0.02)	(0.02)	(0.02)	(0.02)
cb_know_cwt_idx	-0.07	-0.03	0.00	-0.05
	(0.08)	(0.04)	(0.06)	(0.07)
_cons	-0.27***	-0.03	0.46***	0.27***
	(0.05)	(0.05)	(0.05)	(0.05)
N	1851	1851	1851	1851
P value	0.01	0.00	0.02	0.06
BH P value	NA	0.01	0.06	0.09

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

Table A.28: Average treatment effects on knowledge of rules governing local security groups (continued)

	(1) Checkpoints prohibited	(2) Only operate in home community	(3) Can perform citizens' arrest
treatment	0.02	-0.01	0.02*
	(0.02)	(0.04)	(0.01)
cb_know_cwt_idx	-0.06	-0.06	0.01
	(0.05)	(0.11)	(0.03)
_cons	-0.01	0.31***	0.90***
	(0.05)	(0.06)	(0.03)
N	1851	1851	1851
P value	0.23	0.74	0.03
BH P value	0.28	0.74	0.06

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

WILLINGNESS TO REPORT CRIMES TO POLICE

Table A.29: Average treatment effects on willingness to report crimes to police

	(1)	(2)	(3)
	Crime resolution index	Would prefer police response for:	
	index	Land disputes?	Violent land disputes?
treatment	0.01	0.03	0.02
	(0.04)	(0.02)	(0.02)
cb_crimeres_idx	0.09	0.02	0.09
	(0.09)	(0.06)	(0.05)
_cons	0.13^{+}	0.52^{***}	0.65^{***}
	(0.07)	(0.04)	(0.05)
N	1851	1851	1851
P value	0.70	0.18	0.42
BH P value	NA	.36	0.53

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

Table A.30: Average treatment effects on willingness to report crimes to police (continued)

	(1)	(2)	(3)
		Would prefer police response for:	
	Burglaries?	Domestic violence?	Armed robbery?
treatment	-0.01	0.03	-0.03
	(0.02)	(0.03)	(0.02)
cb_crimeres_idx	0.03	0.03	0.02
	(0.06)	(0.06)	(0.04)
_cons	0.79^{***}	0.40^{***}	0.88^{***}
	(0.05)	(0.05)	(0.03)
N	1851	1851	1851
P value	0.81	0.22	0.10
BH P value	0.81	0.36	0.36

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

CRIME TIPS & INFORMATION SHARING

Table A.31: Average treatment effects on crime tips and information sharing

	(1) Crime tips & info sharing index	(2) Reported suspicious activity past 6 months	(3) Helped police find suspect past 6 months	(4) Gave info for police investigation past 6 months	(5) Provided testimony past 6 months
treatment	-0.03	-0.01	-0.00	-0.02	0.00
	(0.04)	(0.01)	(0.01)	(0.01)	(0.01)
cb_crime_tips_idx	0.03	0.06*	-0.01	0.01	-0.02
	(0.08)	(0.03)	(0.03)	(0.03)	(0.02)
_cons	-0.26***	0.02	-0.01	0.01	0.03
	(0.06)	(0.03)	(0.02)	(0.03)	(0.02)
N	1847	1851	1849	1849	1851
P value	0.47	0.37	0.81	0.19	0.88
BH P value	NA	0.74	0.88	0.74	0.88

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

WILLINGNESS TO REPORT POLICE MISCONDUCT

Table A.32: Average treatment effects on willingness to report police misconduct

	(1) Willingness to report police misconduct	(2) Would report illegal checkpoint	(3) Would report drunk officer	(4) Would report police beating
treatment	0.00	0.04^{+}	-0.00	0.02
	(0.04)	(0.02)	(0.02)	(0.02)
cb_police_abuse_report_idx	0.03	0.03	0.06	0.00
	(0.08)	(0.05)	(0.06)	(0.04)
_cons	-0.04	0.65***	0.66***	0.78***
	(0.08)	(0.05)	(0.06)	(0.04)
N	1823	1849	1849	1851
P value	0.92	0.07	0.84	0.34
BH P value	NA	0.21	0.84	0.51

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

INCIDENCE OF CRIME

Table A.33: Average treatment effects on crime

	(1) Total # of crimes reported	(2) # of armed robberies	(3) # of burglaries	(4) # of aggravated assaults
treatment	-0.30	-0.02	-0.06	0.05
	(0.52)	(0.10)	(0.20)	(0.12)
cb_crime_num_lbr	0.41	-0.03	0.15	-0.03
	(0.31)	(0.03)	(0.10)	(0.05)
_cons	3.55^{**}	0.29^{*}	0.86	0.47^{+}
	(1.29)	(0.14)	(0.53)	(0.26)
N	1778	1846	1845	1845
P value	0.57	0.85	0.76	0.70
BH P value	NA	0.94	0.94	0.94

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

Table A.34: Average treatment effects on crime (continued)

	(1) # of simple assaults	(2) # of sexual violence	(3) # of domestic violence	(4) # of domestic verbal abuse
treatment	-0.07	-0.01	-0.08	-0.29
	(0.12)	(0.02)	(0.10)	(0.18)
cb_crime_num_lbr	0.03	0.01	0.12^{*}	0.22
	(0.05)	(0.01)	(0.06)	(0.14)
_cons	0.35	0.09^{*}	0.70^{**}	0.58
	(0.28)	(0.04)	(0.25)	(0.44)
N	1844	1850	1848	1837
P value	0.94	0.94	0.94	0.96

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

Table A.35: Average treatment effects on crime (continued)

	(1) # of violent land disputes	(2) # of non violent land disputes	(3) # of murders	(4) # of child abuse
treatment	0.00	0.02	0.02	-0.03
	(0.05)	(0.05)	(0.02)	(0.16)
cb_crime_num_lbr	-0.02	-0.00	0.00	0.01
	(0.02)	(0.02)	(0.01)	(0.06)
_cons	-0.05	-0.11	0.00	0.30
	(0.09)	(0.13)	(0.02)	(0.35)
N	1846	1817	1851	1848
P value	0.96	0.61	0.40	0.83
BH P value	0.96	0.94	0.94	0.94

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

PERCEPTIONS OF SECURITY

Table A.36: Average treatment effects on perceptions of security

	(1) Perceptions of security	(2) Fears violent crime	(3) Fears non violent crime	(4) Fears walking at night	(5) Fears home invasions
treatment	-0.01	-0.01	-0.00	-0.00	0.00
	(0.06)	(0.03)	(0.03)	(0.03)	(0.03)
cb_future_insecurity_idx	-0.15	0.10^{+}	0.08	0.12^{+}	0.02
	(0.13)	(0.05)	(0.05)	(0.06)	(0.04)
_cons	0.07	0.58^{***}	0.61^{***}	0.28^{***}	0.23^{***}
	(0.10)	(0.05)	(0.05)	(0.07)	(0.04)
N	1850	1851	1851	1851	1851
P value	0.86	0.86	0.94	0.88	0.93
BH P value	NA	0.94	0.94	0.94	0.94

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

Table A.37: Average treatment effects on perceptions of security (continued)

	(1) House boundaries secure	(2) House items secure	(3) Motorbike secure	(4) Generator outside
treatment	-0.01	0.02	0.04^{+}	0.03
	(0.04)	(0.03)	(0.02)	(0.02)
cb_future_insecurity_idx	0.02	0.00	-0.05	-0.07
	(0.06)	(0.06)	(0.06)	(0.05)
_cons	0.48^{***}	0.42^{***}	0.09^{*}	0.08^{*}
	(0.05)	(0.05)	(0.04)	(0.03)
N	1851	1851	1851	1851
P value	0.83	0.59	0.07	0.18
BH P value	0.94	0.94	0.56	0.72

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

SATISFACTION WITH POLICE PERFORMANCE

Table A.38: Average treatment effects on satisfaction with police performance

	(1) Satisfaction w. police performance index	(2) Trusts police	(3) Satisfied w. police performance
treatment	0.05	0.02	0.03
	(0.05)	(0.02)	(0.02)
cb_satis_idx	-0.08	-0.06	0.02
	(0.11)	(0.05)	(0.05)
_cons	0.60***	0.74***	0.79***
	(0.11)	(0.05)	(0.05)
N	1839	1851	1851
P value	0.28	0.41	0.19
BH P value	NA	0.41	0.38

Weighted least squares with block fixed effects, following Section 5.3 of the main text. Cluster robust standard errors in parenthesis, clustered by community. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$.

A.11 Effects on crime reporting using LNP data

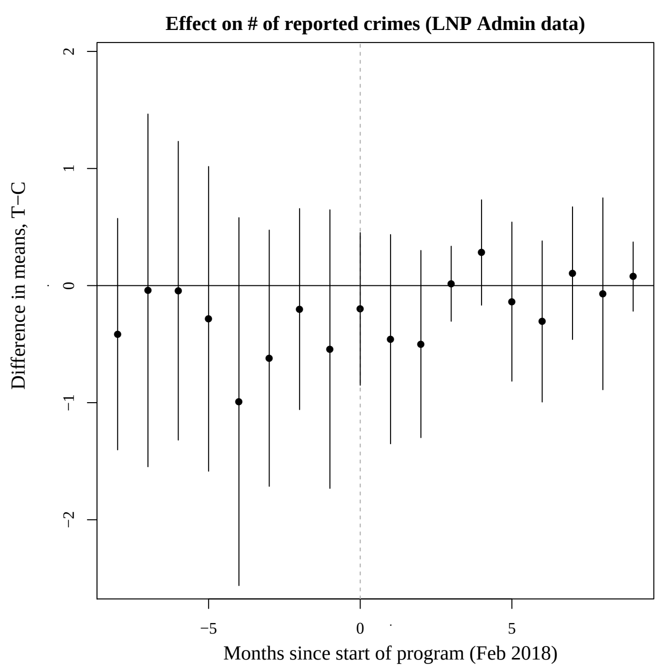
I also collect and analyze administrative crime data from the LNP (see Section A.13 for a full list of administrative crime variables). However, these data are not used to directly test any of the 15 hypotheses, because in addition to being very noisy, they reflect both the incidence of crime and the rate of crime reporting, making it difficult to interpret the results. Instead, I interpret these impacts cautiously and with reference to effects on crime outcomes in the survey, viewing them as corroborative rather than conclusive.

This analysis is conducted at the community-level using weighted least squares regression to account for variation in the probability of assignment to treatment/control across blocks. Specifically, I estimate:

$$y_{vb} = \beta_0 + \beta_1 T_{vb} + \gamma_b + \epsilon_{vb} \tag{1}$$

where y_{vb} indicates the outcome for community v in randomization block (i.e. police zone) b, γ_b denote randomization block fixed effects, and T_{vb} denotes community-level treatment assignment. Results of this analysis are reported below.

Figure A.2: Effect on crime reporting using LNP data



A.12 Results for analyses included in the pre-analysis plan but excluded from the paper

As part of its participation in the EGAP Metaketa IV Initiative, this study pre-specified that it would report impacts on two secondary outcomes: trust in government and trust within communities. Results on these outcomes were excluded from the main paper but are included in Table A.39, below.

Table A.39: Average treatment effects on secondary outcomes excluded from main paper

	Trust in government		Trust in other community members	
treatment	-0.04	-0.08	-0.06	-0.07
	(0.06)	(0.06)	(0.05)	(0.05)
Covariates	N	N	N	N
Ctrl mean	2.87	2.87	2.29	2.29
P-value	0.50	0.18	0.26	0.21
N	1805	1805	1840	1840

Robust standard errors in parenthesis, clustered by community.
$^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < .01$, *** $p < .001$

Patterns of crime reporting in treatment and control communities

In my pre-analysis plan, I noted that crime reporting is measured in the crime victimization module of the survey. Because crime-reporting is post-treatment, differences in patterns of reporting crimes to the police across treatment and control communities does not have a causal interpretation. However, in the pre-analysis plan, I noted that I would document patterns of crime reporting across treatment and control communities, according to the following procedure: for each incident of crime, respondents were asked whether it was reported to the police, local leaders, or nowhere at all. To analyze these data, I employ a crime-level specification given by

$$y_{civb} = \alpha + \beta T_{vb} + \gamma_b + \mathbf{X}_{ivb}\theta + e_{civb} \tag{2}$$

where y_{civb} indicates whether crime c reported by individual i in community v of randomization block b was reported to the police or to local leaders. T_{vb} denotes community-level treatment assignment, $\mathbf{X}_{ivb}$ denotes the individual-level controls for age, gender household size, religion, education, and literacy, and γ_b denotes block fixed effects. Standard errors are clustered at the community level.

Table A.40 reports the results of this analysis. There were no statistically significant differences in reporing crimes to either the police, local leaders, or nowhere at all.

Table A.40: Average treatment effects on crime reporting

	Crime reported to:		
	Nowhere	Police	Leaders
All crimes			
Treatment	-0.01	-0.02	0.02
	(0.02)	(0.02)	(0.02)
Ctrl mean	0.55	0.28	0.11
N	3763	3763	3763
Violent crimes			
Treatment	-0.04	-0.01	0.04
	(0.02)	(0.03)	(0.02)
Ctrl mean	0.50	0.33	0.10
N	1770	1770	1770
Non-violent crimes			
Treatment	0.03	-0.04	0.01
	(0.03)	(0.02)	(0.02)
Ctrl mean	0.60	0.23	0.12
N	1993	1993	1993

Robust standard errors in parenthesis, clustered by community. * $p < 0.05$, ** $p < .01$, *** $p < .001$

A.13 MEASUREMENT OF MECHANISMS

FAMILIARITY WITH THE POLICE

I measure *familiarity with the police* using the following nine questions:

1. The police have special officers responsible for dealing with issues of sexual assault and child abuse. True or False?

2. IF YES: What is the name of the unit to which these officers are assigned? [ANSWER: Women and Child Protection Services (WACPS). ENUMERATOR: IS THE RESPONDENT CORRECT?]

3. The police have a special unit responsible for handling citizen complaints about police misconduct or abuse. True or false?

4. What is the name of this unit? [ANSWER: Professional Standards Division. ENUMERATOR: IS THE RESPONDENT CORRECT?]

5. The police have a special unit responsible for investigating crimes and gathering evidence. What is the name of this unit? [ANSWER: Crime services division. ENUMERATOR: IS THE RESPONDENT CORRECT?]

6. Do you know where the nearest police station is? [ENUMERATOR: IS RESPONDENT CORRECT?]

7. Do you know the name of the COMMANDER at the police station that is nearest to you? [ENUMERATOR: IS RESPONDENT CORRECT?]

8. Do you know the name of any police officer at the police station that is nearest to you?
9. Do you know the PHONE NUMBER of any police officer at the police station that is nearest to you?

For each question, I construct a dummy variable indicating whether the respondent answered affirmatively or correctly, as appropriate. Endline responses are standardized by the baseline mean and standard deviation; the composite index *know_pol_idx* takes the average of these nine standardized variables.

KNOWLEDGE OF THE CRIMINAL JUSTICE SYSTEM

To measure *knowledge of the criminal justice system*, I ask the following nine questions:

1. *know_law_suspect*: If you see a dead body lying in the street and you report it to the police, Liberian law says the police must hold you as a suspect. True or false?
2. *know_law_lawyer*: If you take your case to court and you don't have money to pay a lawyer, Liberian law says the government must provide a lawyer for you. True or false?
3. *know_law_fees*: If you take a case to the police, Liberian law says the police can charge a fee to register the case. True or false?
4. *know_report_station*: Do you know where the nearest police station is? [ENUMERATOR: IS RESPONDENT CORRECT?]
5. *know_law_statrape*: If a man does man-woman business with a woman under age 18, Liberian law says that is rape, even if the woman consents. True or false?
6. *know_law_childsup*: According to Liberian law, a man does not have to provide for his children if he never married the mother and they are seperated. True or false?
7. *know_law_habeasc*: If the police put someone in jail and no one comes to carry a case against that person, Liberian law says the police have to let him go free. True or false?
8. *know_law_bondfee*: If you take a case to court, Liberian law says the Judge can charge you a fee before he can hear the case. True or False?
9. *know_law_complain*: If you report a serious crime to the police like murder or rape and the police fail to take it seriously or investigate, Liberian law says you have the right to file a complaint against the police. True or False?

Questions 1 - 4 were harmonized across all studies in Blair et al. (2021)'s meta-analysis; Questions 5 - 9 are unique to Liberia and address issues that were to be emphasized in the intervention. The meta-analysis will report results on the harmonized index, *know_idx*, which will be constructed from questions 1 - 4; this study will report results on the more-extensive, Liberia-specific index, *know_idx_lbr*, which will be constructed from all 9 questions. For each question, I construct a dummy variable indicating whether the respondent answered correctly; endline responses will be standardized by the baseline mean and standard deviation, and each of the composite indices will take the average of the relevant standardized variables.

PERCEPTIONS OF POLICE INTENTIONS

To measure *perceptions of police intentions*, the survey included the following ten questions:

1. *polint_corrupt*: The police are corrupt or eating money. Agree or disagree?
2. *polint_quality*: The police treat all citizens equally. Agree or disagree?
3. *polcaseserious*: Imagine someone is the victim of armed robbery in your community. The police will take the case seriously and investigate. Agree or disagree?
4. *polcasefair*: Imagine someone is the victim of armed robbery in your community. The police will be fair to all sides in the investigation. Agree or disagree?
5. *pol_care*: The police care about the safety and well-being of people in my community. Agree or disagree?
6. *polint_digresp*: Police treat citizens with dignity and respect. Agree or disagree?
7. *polint_decfact*: Police make their decisions based upon facts, not their personal biases or opinions. Agree or disagree?
8. *polcaserespect*: Imagine someone is the victim of armed robbery in your community. The police will treat the victim with dignity and respect. Agree or disagree?

Questions 1 - 4 were harmonized across all studies participating in Blair et al. (2021)'s meta-analysis; Questions 5 - 8 are unique to Liberia. Accordingly, the Meta-analysis reports results on the harmonized index, *intentions_idx*, which is constructed from questions 1 - 4; this study reports results on the more-extensive, Liberia-specific index, *intentions_idx_lbr* , which is constructed from all 8 questions.

Response options range from “strongly agree” to “strongly disagree” including a neutral “neither agree nor disagree” option, in addition to options for “Do not know” and “Refuse to answer.” Responses are coded on an ordinal scale ranging from 0 (“strongly disagree”) to 4 (“strongly agree”), with “Do not know” and “Refuse to answer” treated as missing. At endline, these variables are standardized by the baseline mean and standard deviation, and each of the composite indices takes the average of the relevant standardized variables.

Perceptions of police capacity

To measure *perceptions of police intentions*, I ask:

- The police have the ability to respond to incidents of crime in a timely manner. Agree or disagree?
- The police have the ability to investigate crimes and gather evidence effectively and professionally. Agree or disagree?

Response options range from “strongly agree” to “strongly disagree” including a neutral “neither agree nor disagree” option, in addition to options for “Do not know” and “Refuse to answer.” Responses are coded on an ordinal scale ranging from 0 (“strongly disagree”) to 4 (“strongly agree”), with “Do not know” and “Refuse to answer” treated as missing. At endline, these variables are standardized by the baseline mean and standard deviation, and the composite index, *police_capacity_idx*, takes the average of the standardized variables.

Perceptions of police responsiveness

To measure *perceptions of police responsiveness*, I ask:

1. *responsive_listen*: The police give people in my community a chance to express their views before making decisions. Agree or disagree?
2. *responsive_act*: The police act upon citizen comments and complaints about security in my community. Agree or disagree?

Question 1 was harmonized across all studies in Blair et al. (2021)’s meta-analysis; Questions 2 was unique to Liberia. The meta-analysis reports results on *responsive_act* alone, while this study reports results on the more-extensive, Liberia-specific index, *pol_responsiveness_lbr* , which is constructed from both questions.

Response options range from "strongly agree" to "strongly disagree" including a neutral "neither agree nor disagree" option, in addition to options for "Do not know" and "Refuse to answer." Responses are coded on an ordinal scale ranging from 0 ("strongly disagree") to 4 ("strongly agree"), with "Do not know" and "Refuse to answer" treated as missing. At endline, these variables are standardized by the baseline mean and standard deviation, and the *pol_responsiveness_lbr* index takes the average of the relevant standardized variables.

NORMS OF CITIZEN COOPERATION WITH POLICE

To measure *norms of citizen cooperation with the police*, I ask:

1. *reportnorm_theft*: If there is a BURGLARY in your community, people can get angry if you take it to the police. Agree or disagree?
2. *reportnorm_abuse*: If a MAN BEATS HIS WIFE in your community, people can get angry if you take it to the police. Agree or disagree?
3. *obeynorm*: You should do what the police tell you to do even when you do not understand the reasons for their decisions. Agree or disagree?
4. *reportnorm_land*: If there is a LAND DISPUTE in your community, people can get angry if you take it to the police. Agree or disagree?
5. *helppolnorm_armedrob*: If a member of the community provides the police with information that helped catch the perpetrator of the [ARMED ROBBERY], other people can get angry with him. Agree or disagree?
6. *helppolnorm_domviol*: If a member of the community provides the police with information that helped catch the perpetrator of the [DOMESTIC VIOLENCE], other people can get angry with him. Agree or disagree?
7. *helppolnorm_moto*: If a member of the community provides the police with information that helped catch the perpetrator of the [MOTORBIKE THEFT], other people can get angry with him. Agree or disagree?
8. *helppolnorm_childabuse*: If a member of the community provides the police with information that helped catch the perpetrator of [CHILDABUSE], other people can get angry with him. Agree or disagree?

Questions 1 - 3 were harmonized across all studies in Blair et al. (2021)'s meta-analysis; Questions 4 - 8 are unique to Liberia. The meta-analysis reports results on the harmonized index, *norm_idx*, which will be constructed from questions 1 - 3; this study reports results on the more-extensive, Liberia-specific index, *norm_idx_lbr*, which is constructed from all 8 questions.

Response options range from "strongly agree" to "strongly disagree" including a neutral "neither agree nor disagree" option, in addition to options for "Do not know" and "Refuse to answer." Responses are coded on an ordinal scale ranging from 0 ("strongly disagree") to 4 ("strongly agree"), with "Do not know" and "Refuse to answer" treated as missing. At endline, these variables are standardized by the baseline mean and standard deviation, and each of the composite indices takes the average of the relevant standardized variables.

SUPPORT FOR MOB VIOLENCE

To measure *support for mob violence*, we ask respondents to judge whether mob violence would be justified in the following three scenarios:

1. Let's say somebody bust into a lady's house and tries to rape her in the night. As he is running away, the community people catch the man and say they want to flog him rather than carry him to the police because they know the police do not have enough manpower to investigate and prepare the case for court. Would you say the actions of the community people are justified, somewhat justified, or not at all justified?
2. Let's say somebody bust into a man's house with a gun, terrorizes his family, and runs away with a his TV and generator. As he is running away, the community people catch the man and want to flog him rather than carry him to the police because they know the police will just release him. Would you say the actions of the community people are justified, somewhat justified, or not at all justified?
3. Let's say somebody busts into another man's home and steals a bag of rice. As he runs away, the community people catch the man and want to flog him rather than carry him to the police because they know the police will not take the case seriously. Would you say the actions of the community people are justified, somewhat justified, or not at all justified?

Response options include "justified", "somewhat justified", and "not at all justified", in addition to options for "Do not know" and "Refuse to answer." Responses will be coded on an ordinal scale ranging from 0 ("Not at all justified ") to 2 ("Justified"), with "Do not know" and "Refuse to answer" coded as missing. At endline, these variables are standardized by the baseline mean and standard deviation, and the composite indices take the average of the three standardized variables.

Knowledge of rules governing local security groups

Knowledge of rules governing local security groups is measured by the following seven questions:

1. *know_cwt_arrest*: Members of the Community Watch Forum may arrest someone provided they carry them to the police and do not physically harm them. True or false?
2. *know_cwt_risk*: If members of the Watch Forum encounter a gang of violent criminals, they are required to engage the criminals even if it puts them at risk. True or false?
3. *know_cwt_checkpoint*: Members of the Community Watch Forum may put up checkpoints if they deem in necessary for security. True or False?
4. *know_cwt_jurisdiction*: Members of the Community Watch Forum may patrol in communities other than their own if they think it is necessary. True or false?
5. *know_cwt_violent*: If a VIOLENT crime is reported to the Community Watch Forum, they are required by law to report it to the police. True or False?
6. *know_cwt_beat*: If a criminal is resisting arrest, the Community Watch Forum has the right to flog him until he can no longer resist. True or False?
7. *know_cwt_cutlass*: Members of the Community Watch Forum have the right to carry cutlasses for protection at night. True or false?

For each question, a dummy variable indicating whether the respondent answered correctly is constructed. Endline responses are standardized by the baseline mean and standard deviation, and the composite index, *know_cwt_idx* takes the average of these seven standardized variables.

Coproduction of security

The degree to which residents *contribute to the coproduction of security* is measured using the following eight questions:

1. In the past month, have you seen or heard of members of your commnunity organizing a security meeting?
2. In the past month, have you attended any security meetings organized by members of your community?

3. In the past month, have you seen or heard of members of your community conducting security patrols at night?
4. In the past month, have you or a member of your family participated in a security patrol organized by members of your community?
5. In the past month, have you or a member of your family donated money, tea, bread, or any other thing to help members of your community conduct security patrols?
6. Does your community currently have an community watch team or community watch forum?
7. (IF YES): Does the watch team/forum conduct nighttime patrols on a regular basis?
8. (IF YES): Does the watch team/forum organize community meetings on a regular basis?

For each question, an indicator variable taking a value of one if the respondent answered affirmatively and zero otherwise is constructed. Endline responses are standardized by the baseline mean and standard deviation, and the composite index, *ca_sec_idx*, takes the average of these eight variables.

POLICE PRESENCE

Police presence is measured using three questions:

1. *compliance_patrol*: About how often do you see police officers patrolling your area on FOOT?
2. *compliance_freq*: About how often do you see police officers patrolling your area while in a vehicle or on a motorbike?
3. *compliance_meeting*: In the past 6 months, have you HEARD ABOUT, SEEN, OR ATTENDED community meetings with police officers in your area?

Responses for each question are standardized by the baseline mean and standard deviation and used to construct *compliance_idx*, which takes the mean of these three standardized variables.

A.14 MEASUREMENT OF PRIMARY OUTCOMES

INCIDENCE OF CRIME

The survey asks the following questions about crime:

1. *armedrob_any*: In the past 6 months, were you or anyone in your family the victim of any ARMED ROBBERY? [ROBBERY WITH ANY KIND OF WEAPON, INCLUDING GUNS, CUTLASSES, STICKS, ETC.]

2. *burglary_any*: In the past 6 months, besides any armed robbery, were you or anyone in your family the victim of BURGLARY or THEFT OF PROPERY? [ROBBERY WITHOUT WEAPON]

3. *aggassault_any*: In the past 6 months, has anyone attacked you or a member of your household WITH A WEAPON? [INCLUDING GUNS, CUTLASSES, STICKS, ETC.]

4. *simpleassault_any*: In the past 6 months, has anyone attacked you or a member of your household WITHOUT a weapon?

5. *sexual_any*: In the past 6 months, have you or anyone in your household been forced or coerced to engage in unwanted sexual activity?

6. *domestic_phys_any*: Apart from what you have told me, in the past 6 months, has anyone in your household ever PHYSICALLY ABUSED you or another person in your household? [INCLUDING PUSHING, SLAPPING, PUNCHING, KICKING, CHOKING, ETC.]

7. *domestic_verbal_any*: Besides any physical abuse, in the past 6 months, has anyone in your household ever been VERBALLY ABUSED by another person in this household? [INCLUDING SHOUTING, CUSSING, THREATS OF ABUSE, ETC.]

8. *land_viol_any*: In the past 6 months, did you or a member of your household have a dispute over land or property that involved THREATS OR VIOLENCE? This include disputes that are still ongoing up to now.

9. *land_nviol_any*: In the past 6 months, did you or a member of your household have a dispute over land or property that did not involve threats or violence? This include disputes that are still ongoing up to now.

10. *other_any*: In the past 6 months, have you or a member of your household been a victim of any OTHER CRIME that we haven't mentioned already?

11. *carmedrob_any*: In the past 6 months, was anyone you know in this community a victim of ARMED ROBBERY? [ROBBERY WITH ANY KIND OF WEAPON, INCLUDING GUNS, CUTLASSES, STICKS, ETC.]

12. *cburglary_any*: In the 6 months, was anyone you know in this community a victim of BURGLARY or THEFT? [ROBBERY WITHOUT WEAPON]

13. *caggassault_any*: In the past 6 months, was anyone you know in this community attacked WITH A WEAPON? [INCLUDING GUNS, CUTLASSES, STICKS, ETC.]

14. *csimpleassault_any*: In the past 6 months, was anyone you know in this community attacked WITHOUT a weapon? [SIMPE ASSAULT]

15. *csexual_any*: In the past 6 months, was anyone you know in this community SEXUALLY ABUSED? [INCLUDING RAPE]

16. *cdomestic_phys_any*: In the 6 months, was anyone you know in this community been PHYSICALLY ABUSED by someone in their own household? [INCLUDING PUSHING, SLAPPING, PUNCHING, KICKING, CHOKING, ETC.]

17. *cdomestic_verbal_any*: In the past 6 months, was anyone you know in this community VERBALLY ABUSED by someone in their own household? [INCLUDING SHOUTING, CUSSING, THREATS OF ABUSE, ETC.]

18. *cland_viol_any*: In the past 6 months, did anyone you know in this community have a LAND DISPUTE over their house land or farm land involving THREATS OR VIOLENCE? This includes disputes that started in the past and have been resovled or ones that are still ongoing up to now.

19. *cland_nviol_any*: In the past 6 months, did anyone you know in this community have a LAND DISPUTE over their house land or farm land that did not involve threats or violence? This includes disputes that started in the past and have been resovled or ones that are still ongoing up to now.

20. *cmurder_any*: In the past 6 months, was anyone you know in this community MURDERED?

21. *cchildabuse_any*: In the past 6 months, was any CHILD you know of in your community a victim of CHILD ABUSE? [INCLUDING LACK OF FOOD, LACK OF SUPPORT, HITTING, BEATING, CHILD LABOR, OR SEXUAL ABUSE?]

22. *cother_any*: In the past 6 months, was anyone you know in your community a victim of any OTHER CRIME that we haven't mentioned already?

For each affirmative response, a follow-up question asks "How many times did this happen in the past 6 months?" Questions 3, 5, 6, 7, 8, 9, 18, 19, and 21 were only asked in Liberia; the remaining questions were asked in Liberia as well as in the other studies participating in Blair et al. (2021)'s meta-analysis.

For the meta-analysis, *violentcrime_num*, denoting the total number of violent crimes a respondent reports occurred in their community in the past 6 months, is constructed from questions on armed robbery, simple assault, aggravated assault, sexual assault, murder, domestic violence, and violent crimes in the "other" category. Questions on burglary and non-violent crimes in the "other" category are used to construct *nonviolentcrime_num*, denoting the total number of non-violent crimes a respondent reports occurred in their community in the past 6 months. Both of these indices are used to construct *crime_num*, the composite index for this cluster, denoting the total number of crimes a respondent reports occurred in their community in the past 6 months.

For this study, I report effects on *crime_num_lbr*, an index denoting the total number of crimes reported across all categories measured in Liberia. In addition, I report effects on each of the following individual categories of crime: burglary, armed robbery, simple assault, aggravated assault, sexual violence, domestic violence, domestic abuse, and land conflict (violent). For each category, a continuous variable is constructed as the sum of i) crimes against the respondent or their family members and ii) crimes that occurred to other people the respondent knows in their community.

CRIME REPORTING

For each affirmative response to the crime questions listed below, a follow-up question asks (with respect to the most recent time the crime occurred): "Where did you report this case?" with answer options: nowhere, police, courts, town leader, elders, community watch group/forum, settled directly with perpetrator, and other.[20] Following the meta-analysis PAP, these questions are used to construct *crime_report_num*, the total number of crimes reported to the police; *violentcrime_report_num*, the total number of violent crimes reported to the police; and *nonviolentcrime_report_num*, the total number of non-violent crimes reported to the police.

The survey also asks the following questions about *support for crime reporting* in hypothetical scenarios:

1. *burglaryres*: If there's a BURGLARY in your community, where do you think the case should be reported, if anywhere?
2. *dviolres*: If a MAN BEAT HIS WOMAN in your community, where do you think the case should be reported, if anywhere?
3. *armedrobres*: If there's an ARMED ROBBERY in your community, where do you think the case should be reported, if anywhere?

[20] For questions about crimes against other people they know in their community, the crime reporting question is phrased "As far as you know, where was this case reported?"

with response options: nowhere, police, courts, town leader, elders, community watch group/forum, and other. For each scenario, I code a dummy variable that takes a value of one for "police" responses and zero otherwise. The index *crimeres_idx* takes the average of these five variables. The primary outcome index for crime reporting, *crime_reporting_idx*, is constructed as the average of: *crime_report_num* and *crimeres_idx*.

Administrative crime data

Administrative data from the LNP is used to construct the following variables:

1. *armedrob_num* : Total number of armed robberies in community in past 6 months
2. *aburglary_num* : Total number of burglaries in community in past 6 months
3. *aaggassault_num* : Total number of aggravated assaults in community in past 6 months
4. *asimpleassault_num* : Total number of simple assaults in community in past 6 months
5. *asexual_num* : Total number of sexual violence incidents in community in past 6 months
6. *adomestic_phys_num* : Total number of domestic violence incidents in community in past 6 months
7. *adomestic_verbal_num* : Total number of verbal domestic violence incidents in community in past 6 months
8. *aland_num* : Total number of armed robberies in land conflicts in community in past 6 months
9. *aland_violent_num* : Total number of violent land conflicts in community in past 6 months
10. *amob_num* : Total number of mob violence incidents community in past 6 months
11. *ariot_num* : Total number of riots in community in past 6 months
12. *amurder_num* : Total number of murders in community in past 6 months
13. *aother_any* : Total number of other crimes in community in past 6 months

Primary outcome indices *acrime_num* and *aviolentcrime_num* are constructed from these variables.

CRIME TIPS AND INFORMATION SHARING

I measure *information sharing* with four questions:

1. *contact_pol_susp_activity*: In the past 12 months, have you ever contacted the police to alert them to suspicious or criminal activity in your community?
2. *give_info_pol_investigation*: In the past 12 months, have you ever given information to the police to assist with an investigation?
3. *contact_pol_find_suspect*: In the past 12 months, have you ever contacted the police to help them find a suspected criminal in your community?
4. *testify_police_investigation*: In the past 12 months, have you ever served as a witness or provided testimony as part of a police investigation?
5. *name_witness*: Suppose there was a robbery in your community. You didn't see the crime occur, but you know someone in your community who did. How likely would you be to give that person's name to the police?
6. *identify_hideout*: Suppose there is a suspect accused of car jacking hiding in your community and the police are looking for him. Let's say you happen to know where that person is hiding. How likely would you be to give that information to the police?
7. *identify_ghetto*: Suppose there is an area of your community where people take drugs and plan petty crimes. How likely would you be to report that information to the police?
8. *guide_police*: Suppose a police officer wants to familiarize himself with your community. How willing would you be to spend a day showing him around your community?
9. *give_testimony*: Suppose you witness a crime in your community and the police ask you to give written testimony at the police station. How likely would you be to go to the station to give written testimony?

Questions 1 - 2 were harmonized across all studies participating in Blair et al. (2021)'s meta-analysis; Questions 3 - 9 are unique to Liberia. The meta-analysis reports results on the harmonized index, *tips_idx*, which is constructed from questions 1 - 2; this study reports results on the more-extensive, Liberia-specific index, *tips_idx_lbr* , which is constructed from all nine questions, with each input variable standardized by the baseline mean and standard deviation.

Willingness to report police abuse

Willingness to report police abuse is measured using three questions:

1. *checkpoint_report*: Suppose you see a group of police officers running an illegal checkpoint just to take money from motorists. How likely would you be to report that situation?

2. *dutydrink_report*: Suppose you see a uniformed police officer drinking alcohol in your community. How likely would you be to report that situation?

3. *policebeating_report*: Suppose you see a group of officers beating someone in your community. How likely would you be to report that situation?

Questions 1 - 2 were included in all studies participating in Blair et al. (2021)'s meta-analysis; Question 3 is unique to Liberia. The Meta-analysis reports results on the harmonized index, *police_abuse_report_idx*, which is constructed from Questions 1 - 2; this study reports results on the more-extensive, Liberia-specific index, *police_abuse_report_idx_lbr*, which is constructed from all three questions, with each input variable standardized by the baseline mean and standard deviation.

Perceptions of security

To measure *perceptions of security*, I ask:

1. *fear_violent*: How worried are you that you or a member of your household will be the victim of a VIOLENT CRIME in the coming year? [INCLUDING ARMED ROBBERY, ASSAULT WITH A WEAPON, ASSAULT WITHOUT A WEAPON, ETC.]

2. *fear_nonviolent*: How worried are you that you or a member of your household will be the victim of a NON-VIOLENT CRIME in the coming year? [INCLUDING BURGLARY, THEFT, ETC.]

3. *feared_walk*: In the past 6 months, how often, if ever, have you or anyone in your family felt unsafe walking in your neighbourhood?

4. *feared_home*: In the past 6 months, how often, if ever, have you or anyone in your family feared crime in your own home?

5. *hssecure*: How sure are you that the boundaries of your house spots are secure? That is, no one can leave from his side to come and sit down on your side?
6. *hsitemssecure*: How sure are you that the valuable items in and around your house are secure? (e.g. generators, phones, computers, TVs, furniture)

Questions 1 - 3 were included in all studies in Blair et al. (2021)'s meta-analysis; Questions 4-6 were unique to Liberia. The meta-analysis reports results on the harmonized index, *future_security_idx*, which is constructed from Questions 1 - 3; this study reports results on the more-extensive, Liberia-specific index, *future_security_idx_lbr*, which is constructed from all six questions, with each input variable standardized by the baseline mean and standard deviation.

SATISFACTION WITH POLICE PERFORMANCE

To measure *satisfaction with police performance*, I ask:

1. *satis_trust*: In general, I trust the police. Agree or disagree?
2. *satis_general*: I am satisfied with the service that police provide. Agree or disagree?

Response options range from "strongly agree" to "strongly disagree" including a neutral "neither agree nor disagree" option, in addition to options for "Do not know" and "Refuse to answer." Responses are coded on an ordinal scale ranging from 0 ("strongly disagree") to 4 ("strongly agree"), with "Do not know" and "Refuse to answer" treated as missing. At endline, these variables are standardized by the baseline mean and standard deviation; the composite index *satis_idx* takes the average of these two standardized variables.

MOB VIOLENCE

To measure mob violence, the survey asked "In the past year, were there any incidents of MOB JUSTICE in your community (i.e. beating of flogging of someone suspected of committing a crime)?", followed by "How many times did this happen in the past year (12 months)?" for affirmative responses. These two questions are used to construct a variable *cmob_num*, denoting the number of mob violence incidents in the past 12 months that the respondent had heard of.

A.15 MEASUREMENT OF SECONDARY OUTCOMES

TRUST IN GOVERNMENT

To measure *Trust in government* (S1) I ask: “Currently, how much do you trust the government?”, with response options: “Not at all”, “2-Just a little”, “Somewhat”, “A lot”, “Do not know”, and “Refuse to answer”. Responses to these questions are standardized by the baseline mean and standard deviation.

COMMUNAL TRUST

To measure *communual trust* (S2), I ask:

1. *trust_community*: Most people in my community can be trusted. Agree or disagree?
2. *days_comm_work*: In the past 30 days, how many days did you spend doing community work, like cleaning dirt or brushing the road?
3. *cgroups*: How many community groups that meet regularly do you participate in?
4. *trust_keys*: Would you be willing to leave one of your neighbors the keys to your home while you went away for the afternoon?
5. *comm_help*: Most members of [COMMUNITY NAME] are willing to help me when I'm in need. Agree or disagree?

Questions 1 was included in all studies participating in the meta-analysis; Questions 2-5 were unique to Liberia. The Meta-analysis reports results on the harmonized index, *trust_community*; this study reports results on a more-extensive, Liberia-specific index, *comm_cohesion*, which is constructed from all five questions, with each input variable standardized by the baseline mean and standard deviation.